

MATHEMATICS BASIC 2026 ALL SETS

CHAPTER 1: REAL NUMBERS

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If the HCF of two positive integers a and b is 1, then their LCM is:

OPTIONS: (A) $a+b$ (B) a (C) b (D) ab

ANSWER: (D) ab

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The number $3 + \sqrt{2}$ is:

OPTIONS: (A) a rational number (B) an irrational number (C) an integer (D) a natural number

ANSWER: (B) an irrational number

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following numbers will not end with 0 for any natural number n ? (A) $4n$ (B) 4^n (C) $3^n + 1$ (D) $10^{(n+1)}$

OPTIONS: (A) $4n$ (B) 4^n (C) $3^n + 1$ (D) $10^{(n+1)}$

ANSWER: (B) 4^n

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The number 2^n , where n is a natural number, cannot end with the digit:

OPTIONS: (A) 4 (B) 6 (C) 2 (D) 0

ANSWER: (D) 0

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If the HCF of two positive integers a and b is 1, then their LCM is:

OPTIONS: (A) $a+b$ (B) a (C) b (D) ab

ANSWER: (D) ab

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The HCF of $2^2 \cdot 3^3$ and $3^2 \cdot 2^3$ is:

OPTIONS: (A) 1 (B) $2 \cdot 3$ (C) $2^2 \cdot 3^2$ (D) $2^3 \cdot 3^3$

ANSWER: (C) $2^2 \cdot 3^2$

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A letter is selected from the letters of the word FEBRUARY. The probability that it is a vowel is:

OPTIONS: (A) $\frac{1}{8}$ (B) $\frac{2}{8}$ (C) $\frac{3}{8}$ (D) $\frac{3}{7}$

ANSWER: (C) $\frac{3}{8}$

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: $(\sqrt{3} - 3)/\sqrt{3}$ is:

OPTIONS: (A) a rational number (B) an irrational number (C) an integer (D) a natural number

ANSWER: (A) a rational number

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The HCF of 3^5 and 5^3 is:

OPTIONS: (A) $3^0 \cdot 5^0$ (B) $3^3 \cdot 5^3$ (C) $3^1 \cdot 5^1$ (D) $3^5 \cdot 5^3$

ANSWER: (A) $3^0 \cdot 5^0$

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The LCM of two consecutive natural numbers p and $p+1$ is:

OPTIONS: (A) p (B) $p^2 + p$ (C) 1 (D) $2p + 1$

ANSWER: (B) $p^2 + p$

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The LCM of the smallest 2-digit number and the smallest 3-digit number is expressed in the form $2^p \cdot 5^q$. The respective values of p and q are:

OPTIONS: (A) 1,2 (B) 1,1 (C) 2,1 (D) 2,2

ANSWER: (D) 2,2

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If the HCF of the smallest odd prime number and the greatest 2-digit number is expressed as $3^m \cdot 11^n$, then the values of m and n respectively are:

OPTIONS: (A) 0,0 (B) 1,0 (C) 1,1 (D) 2,1

ANSWER: (B) 1,0

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The HCF of 2^3 and 3^2 is:

OPTIONS: (A) $2^0 \cdot 3^0$ (B) $2^1 \cdot 3^1$ (C) $2^3 \cdot 3^2$ (D) $2^2 \cdot 3^3$

ANSWER: (A) $2^0 \cdot 3^0$

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The number 3^n , where n is a natural number, cannot end with the digit:

OPTIONS: (A) 3 (B) 5 (C) 7 (D) 9

ANSWER: (B) 5

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The LCM of two consecutive natural numbers p and $p+1$ is:

OPTIONS: (A) p (B) $p^2 + p$ (C) 1 (D) $2p + 1$

ANSWER: (B) $p^2 + p$

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: $(2 + \sqrt{2})^2$ is:

OPTIONS: (A) a rational number (B) an irrational number (C) an integer (D) a natural number

ANSWER: (B) an irrational number

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The HCF of $3^7 \cdot 7^3$ and $3^3 \cdot 7^7$ is:

OPTIONS: (A) 1 (B) $3 \cdot 7$ (C) $3^3 \cdot 7^3$ (D) $3^7 \cdot 7^7$

ANSWER: (C) $3^3 \cdot 7^3$

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following numbers will not end with 0 for any natural number n ? (A) $4n$ (B) 4^n (C) $3^n + 1$ (D) $10^n + 1$

OPTIONS: (A) $4n$ (B) 4^n (C) $3^n + 1$ (D) $10^n + 1$

ANSWER: (B) 4^n

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If the number a^n , where n is a natural number, always ends with digit a , then the possible value of ' a ' is:

OPTIONS: (A) 2 (B) 4 (C) 6 (D) 8

ANSWER: (C) 6

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find the H.C.F. and L.C.M. of 1530 and 2040.

OPTIONS: (A) HCF = 510, LCM = 6120 (B) HCF = 510, LCM = 1020 (C) HCF = 510, LCM = 3060 (D) HCF = 1020, LCM = 6120

ANSWER: (A) HCF = 510, LCM = 6120

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find the H.C.F. and L.C.M. of 408 and 312.

OPTIONS: (A) HCF = 24, LCM = 5304 (B) HCF = 24, LCM = 2652 (C) HCF = 12, LCM = 5304 (D) HCF = 24, LCM = 3120

ANSWER: (A) HCF = 24, LCM = 5304

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If H.C.F. of $135x^2$ and $189x^3$ is 108, then find the value of x .

OPTIONS: (A) 1 (B) 2 (C) 3 (D) 4

ANSWER: (B) 2

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CHAPTER: Real Numbers

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that $\sqrt{5}$ is an irrational number.

OPTIONS: (A) $\sqrt{5}$ is rational because it can be expressed as a/b (B) $\sqrt{5}$ is irrational because 5 divides both a and b leading to contradiction (C) $\sqrt{5}$ is rational because 5 is a perfect square (D) $\sqrt{5}$ is irrational because it is a decimal

ANSWER: (B) $\sqrt{5}$ is irrational because 5 divides both a and b leading to contradiction

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that $\sqrt{3}$ is an irrational number.

OPTIONS: (A) $\sqrt{3}$ is rational because it can be expressed as a/b (B) $\sqrt{3}$ is irrational because 3 divides both a and b leading to contradiction (C) $\sqrt{3}$ is rational because 3 is a perfect square (D) $\sqrt{3}$ is irrational because it is a decimal

ANSWER: (B) $\sqrt{3}$ is irrational because 3 divides both a and b leading to contradiction

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that $\sqrt{2}$ is an irrational number.

OPTIONS: (A) $\sqrt{2}$ is rational because it can be expressed as a/b (B) $\sqrt{2}$ is irrational because 2 divides both a and b leading to contradiction (C) $\sqrt{2}$ is rational because 2 is a perfect square (D) $\sqrt{2}$ is irrational because it is a decimal

ANSWER: (B) $\sqrt{2}$ is irrational because 2 divides both a and b leading to contradiction

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: State the "Fundamental Theorem of Arithmetic" and use it to find LCM of 36 and 54.

OPTIONS: (A) LCM = 36 (B) LCM = 54 (C) LCM = 108 (D) LCM = 72

ANSWER: (C) LCM = 108

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find which among the following numbers a , b and c is/are composite numbers. $a = 7 \times 11 \times 13 + 13$, $b = 6 \times 5 \times 4 + 4$, $c = 7 \times 13 + 6$

OPTIONS: (A) a and b only (B) a and c only (C) b and c only (D) a , b and c are all composite

ANSWER: (D) a , b and c are all composite

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: (a) Prove that $3+2\sqrt{5}$ is an irrational number.

OPTIONS: (A) It is rational because it can be expressed as a/b (B) It is irrational because $\sqrt{5}$ is irrational and sum of rational and irrational is irrational (C) It is rational because 3 is rational (D) It is irrational because it is a decimal

ANSWER: (B) It is irrational because $\sqrt{5}$ is irrational and sum of rational and irrational is irrational

YEAR: 2026

CHAPTER: Real Numbers

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Prove that $2-5\sqrt{3}$ is an irrational number.

OPTIONS: (A) It is rational because it can be expressed as a/b (B) It is irrational because $\sqrt{3}$ is irrational and sum of rational and irrational is irrational (C) It is rational because 2 is rational (D) It is irrational because it is a decimal

ANSWER: (B) It is irrational because $\sqrt{3}$ is irrational and sum of rational and irrational is irrational

CHAPTER 2: POLYNOMIALS

YEAR: 2026

CHAPTER: Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: One zero of a quadratic polynomial is twice the other. If the sum of zeroes is -6, find the polynomial.

OPTIONS: (A) $x^2 + 6x + 8$ (B) $x^2 - 6x + 8$ (C) $x^2 + 6x - 8$ (D) $x^2 - 6x - 8$

ANSWER: (A) $x^2 + 6x + 8$

YEAR: 2026

CHAPTER: Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If one zero of the polynomial $x^2 - 5x - c$ is -1, find the value of c. Also, find the other zero.

OPTIONS: (A) $c = 6$, other zero = 6 (B) $c = -6$, other zero = -6 (C) $c = 6$, other zero = -6 (D) $c = -6$, other zero = 6

ANSWER: (A) $c = 6$, other zero = 6

YEAR: 2026

CHAPTER: Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find a quadratic polynomial whose sum and product of zeroes are 0 and -9, respectively. Also, find the zeroes.

OPTIONS: (A) $x^2 - 9$, zeroes = 3, -3 (B) $x^2 + 9$, zeroes = 3i, -3i (C) $x^2 - 9$, zeroes = 0, 9 (D) $x^2 + 9$, zeroes = 0, -9

ANSWER: (A) $x^2 - 9$, zeroes = 3, -3

YEAR: 2026

CHAPTER: Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: If α and β are zeroes of the polynomial $p(x) = 5x^2 - 7x - 3$, then form a quadratic polynomial whose zeroes are $2/\alpha$ and $2/\beta$.

OPTIONS: (A) $3x^2 + 14x - 20$ (B) $3x^2 - 14x - 20$ (C) $3x^2 + 14x + 20$ (D) $3x^2 - 14x + 20$

ANSWER: (A) $3x^2 + 14x - 20$

YEAR: 2026

CHAPTER: Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the zeroes of the polynomial $p(x) = 3x^2 + 7x - 20$ and verify the relationship between its zeroes and the coefficients.

OPTIONS: (A) Zeroes = 4, $-\frac{5}{3}$; sum = $\frac{7}{3}$, product = $-\frac{20}{3}$ (B) Zeroes = -4, $\frac{5}{3}$; sum = $-\frac{7}{3}$, product = $-\frac{20}{3}$ (C) Zeroes = 4, $\frac{5}{3}$; sum = $\frac{7}{3}$, product = $\frac{20}{3}$ (D) Zeroes = -4, $-\frac{5}{3}$; sum = $-\frac{7}{3}$, product = $\frac{20}{3}$

ANSWER: (B) Zeroes = -4, $\frac{5}{3}$; sum = $-\frac{7}{3}$, product = $-\frac{20}{3}$

YEAR: 2026

CHAPTER: Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find a quadratic polynomial, sum and product of whose zeroes are 5 and -6 respectively.

OPTIONS: (A) $x^2 - 5x - 6$ (B) $x^2 + 5x - 6$ (C) $x^2 - 5x + 6$ (D) $x^2 + 5x + 6$

ANSWER: (A) $x^2 - 5x - 6$

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CHAPTER: Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If one zero of the polynomial $x^2 - 5x - c$ is -1, find the value of c and the other zero.

OPTIONS: (A) $c = 6$, other zero = 6 (B) $c = -6$, other zero = -6 (C) $c = 6$, other zero = -6 (D) $c = -6$, other zero = 6

ANSWER: (A) $c = 6$, other zero = 6

YEAR: 2026

CHAPTER: Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Determine a quadratic polynomial, sum and product of whose zeroes are -10 and 24 respectively. Also, determine the zeroes.

OPTIONS: (A) $x^2 - 10x + 24$, zeroes = 6, 4 (B) $x^2 + 10x + 24$, zeroes = -6, -4 (C) $x^2 - 10x - 24$, zeroes = 12, -2 (D) $x^2 + 10x - 24$, zeroes = -12, 2

ANSWER: (B) $x^2 + 10x + 24$, zeroes = -6, -4

YEAR: 2026

CHAPTER: Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the zeroes of the polynomial $p(x) = 3x^2 - 2x - 1$ and verify the relationship between the zeroes and the coefficients.

OPTIONS: (A) Zeroes = 1, $-1/3$; sum = $2/3$, product = $-1/3$ (B) Zeroes = -1, $1/3$; sum = $-2/3$, product = $-1/3$ (C) Zeroes = 1, $1/3$; sum = $4/3$, product = $1/3$ (D) Zeroes = -1, $-1/3$; sum = $-4/3$, product = $1/3$

ANSWER: (A) Zeroes = 1, $-1/3$; sum = $2/3$, product = $-1/3$

YEAR: 2026

CHAPTER: Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Verify the relationship between the zeroes and the coefficients of the quadratic polynomial $25x^2 - 16$.

OPTIONS: (A) Zeroes = $4/5$, $-4/5$; sum = 0, product = $-16/25$ (B) Zeroes = $4/5$, $-4/5$; sum = 0, product = $16/25$ (C) Zeroes = $5/4$, $-5/4$; sum = 0, product = $-25/16$ (D) Zeroes = $5/4$, $-5/4$; sum = 0, product = $25/16$

ANSWER: (A) Zeroes = $4/5$, $-4/5$; sum = 0, product = $-16/25$

YEAR: 2026

CHAPTER: Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the zeroes of the polynomial $p(x) = 2x^2 + 5x + 2$ and verify the relationship between zeroes and coefficients.

OPTIONS: (A) Zeroes = $-1/2$, -2; sum = $-5/2$, product = 1 (B) Zeroes = $1/2$, 2; sum = $5/2$, product = 1 (C) Zeroes = $-1/2$, -2; sum = $-5/2$, product = 1 (D) Zeroes = $1/2$, 2; sum = $5/2$, product = -1

ANSWER: (C) Zeroes = $-1/2$, -2; sum = $-5/2$, product = 1

YEAR: 2026

CHAPTER: Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Verify the relationship between the zeroes and the coefficients of the quadratic polynomial $9x^2 - 25$.

OPTIONS: (A) Zeroes = $5/3$, $-5/3$; sum = 0, product = $-25/9$ (B) Zeroes = $5/3$, $-5/3$; sum = 0, product = $25/9$ (C) Zeroes = $3/5$, $-3/5$; sum = 0, product = $-9/25$ (D) Zeroes = $3/5$, $-3/5$; sum = 0, product = $9/25$

ANSWER: (A) Zeroes = $5/3$, $-5/3$; sum = 0, product = $-25/9$

YEAR: 2026

CHAPTER: Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the zeroes of the polynomial $q(x) = 6x^2 - 5x - 1$ and verify the relationship between its zeroes and coefficients.

OPTIONS: (A) Zeroes = 1, $-1/6$; sum = $5/6$, product = $-1/6$ (B) Zeroes = -1 , $1/6$; sum = $-5/6$, product = $-1/6$ (C) Zeroes = 1, $1/6$; sum = $7/6$, product = $1/6$ (D) Zeroes = -1 , $-1/6$; sum = $-7/6$, product = $1/6$

ANSWER: (A) Zeroes = 1, $-1/6$; sum = $5/6$, product = $-1/6$

YEAR: 2026

CHAPTER: Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Verify the relationship between the zeroes and the coefficients of the polynomial $p(x) = 3x^2 - 5x$.

OPTIONS: (A) Zeroes = 0, $5/3$; sum = $5/3$, product = 0 (B) Zeroes = 0, $-5/3$; sum = $-5/3$, product = 0 (C) Zeroes = $5/3$, $-5/3$; sum = 0, product = $-25/9$ (D) Zeroes = 0, $5/3$; sum = $5/3$, product = $5/3$

ANSWER: (A) Zeroes = 0, $5/3$; sum = $5/3$, product = 0

YEAR: 2026

CHAPTER: Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the zeroes of the polynomial $9s^2 - 6s + 1$ and verify the relationship between the zeroes and the coefficients.

OPTIONS: (A) Zeroes = $1/3$, $1/3$; sum = $2/3$, product = $1/9$ (B) Zeroes = $-1/3$, $-1/3$; sum = $-2/3$, product = $1/9$ (C) Zeroes = $1/3$, $1/3$; sum = $2/3$, product = $-1/9$ (D) Zeroes = $-1/3$, $-1/3$; sum = $-2/3$, product = $-1/9$

ANSWER: (A) Zeroes = $1/3$, $1/3$; sum = $2/3$, product = $1/9$

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CHAPTER: Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the zeroes of the polynomial $4x^2 + 4x + 1$ and verify the relationship between zeroes and coefficients.

OPTIONS: (A) Zeroes = $-1/2, -1/2$; sum = -1 , product = $1/4$ (B) Zeroes = $1/2, 1/2$; sum = 1 , product = $1/4$ (C) Zeroes = $-1/2, -1/2$; sum = -1 , product = $-1/4$ (D) Zeroes = $1/2, 1/2$; sum = 1 , product = $-1/4$

ANSWER: (A) Zeroes = $-1/2, -1/2$; sum = -1 , product = $1/4$

YEAR: 2026

CHAPTER: Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find the zeroes of the polynomial $25a^2 - 10a + 1$ and verify the relationship between the zeroes and coefficients.

OPTIONS: (A) Zeroes = $1/5, 1/5$; sum = $2/5$, product = $1/25$ (B) Zeroes = $-1/5, -1/5$; sum = $-2/5$, product = $1/25$ (C) Zeroes = $1/5, 1/5$; sum = $2/5$, product = $-1/25$ (D) Zeroes = $-1/5, -1/5$; sum = $-2/5$, product = $-1/25$

ANSWER: (A) Zeroes = $1/5, 1/5$; sum = $2/5$, product = $1/25$

YEAR: 2026

CHAPTER: Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If α and β are the zeroes of polynomial $p(x) = 6x^2 - 5x - 3$, then find the value of $1/\alpha + 1/\beta$.

OPTIONS: (A) $-5/3$ (B) $5/3$ (C) $-3/5$ (D) $3/5$

ANSWER: (A) $-5/3$

YEAR: 2026

CHAPTER: Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: One zero of the polynomial $4x^2 - 12x + (2k + 1)$ is five times the other. Find the value of k .

OPTIONS: (A) 2 (B) 3 (C) 4 (D) 5

ANSWER: (A) 2

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CHAPTER: Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If α and β are the zeroes of polynomial $p(x) = -9x^2 - 6x + 1$, find the value of $\alpha^2 + \beta^2$.

OPTIONS: (A) $2/3$ (B) $-2/3$ (C) $4/9$ (D) $-4/9$

ANSWER: (A) $\frac{2}{3}$

YEAR: 2026

CHAPTER: Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Form a quadratic polynomial whose zeroes are twice the zeroes of polynomial $p(x) = x^2 - 3x - 5$.

OPTIONS: (A) $x^2 - 6x - 20$ (B) $x^2 + 6x - 20$ (C) $x^2 - 6x + 20$ (D) $x^2 + 6x + 20$

ANSWER: (A) $x^2 - 6x - 20$

YEAR: 2026

CHAPTER: Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: (a) Find the zeroes of the polynomial $p(x) = 4x^2 - 8x + 3$ and verify the relationship between its zeroes and coefficients.

OPTIONS: (A) Zeroes = $\frac{3}{2}$, $\frac{1}{2}$; sum = 2, product = $\frac{3}{4}$ (B) Zeroes = $-\frac{3}{2}$, $-\frac{1}{2}$; sum = -2, product = $\frac{3}{4}$ (C) Zeroes = $\frac{3}{2}$, $\frac{1}{2}$; sum = 2, product = $-\frac{3}{4}$ (D) Zeroes = $-\frac{3}{2}$, $-\frac{1}{2}$; sum = -2, product = $-\frac{3}{4}$

ANSWER: (A) Zeroes = $\frac{3}{2}$, $\frac{1}{2}$; sum = 2, product = $\frac{3}{4}$

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CHAPTER: Polynomials

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: (b) Form a polynomial whose zeroes are α^2 and β^2 , where α and β are zeroes of the polynomial $p(x) = x^2 - 3\sqrt{2}x + 4$.

OPTIONS: (A) $x^2 - 10x + 16$ (B) $x^2 + 10x + 16$ (C) $x^2 - 10x - 16$ (D) $x^2 + 10x - 16$

ANSWER: (A) $x^2 - 10x + 16$

YEAR: 2026

CHAPTER: Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Form a quadratic polynomial whose sum and product of the zeroes are $\frac{1}{2}$ and $-\frac{1}{9}$ respectively. Hence, find the zeroes.

OPTIONS: (A) $18x^2 - 9x - 2$; zeroes = $-\frac{1}{6}$, $\frac{2}{3}$ (B) $18x^2 + 9x - 2$; zeroes = $\frac{1}{6}$, $-\frac{2}{3}$ (C) $18x^2 - 9x + 2$; zeroes = $\frac{1}{6}$, $\frac{2}{3}$ (D) $18x^2 + 9x + 2$; zeroes = $-\frac{1}{6}$, $-\frac{2}{3}$

ANSWER: (A) $18x^2 - 9x - 2$; zeroes = $-\frac{1}{6}$, $\frac{2}{3}$

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CHAPTER: Polynomials

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If α and β are zeroes of the polynomial $x^2 - 6x + 7$, then find the value of $4(1/\alpha^2 + 1/\beta^2)$.

OPTIONS: (A) 88/49 (B) 88/7 (C) 44/49 (D) 44/7

ANSWER: (A) 88/49

YEAR: 2026

CHAPTER: Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of k for which sum of the zeroes of the polynomial $p(x) = 3x^2 - kx + 6$ is 2, is:

OPTIONS: (A) 2 (B) -6 (C) -2 (D) 6

ANSWER: (D) 6

YEAR: 2026

CHAPTER: Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If the zeroes of the polynomial $p(x) = 2x^2 - 7x + 6$ are α and β , then the value of $1/\alpha + 1/\beta$ is:

OPTIONS: (A) 7/2 (B) 6/7 (C) -7/6 (D) 7/6

ANSWER: (D) 7/6

YEAR: 2026

CHAPTER: Polynomials

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of k for which sum of the zeroes of the polynomial $p(x) = 3x^2 - kx + 6$ is 2, is:

OPTIONS: (A) 2 (B) -6 (C) -2 (D) 6

ANSWER: (D) 6

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CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

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YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following system of equations has a unique solution? (A) $x = 0$, $x = 1$
(B) $x + y = 0$, $2x + 2y = 0$ (C) $x + y = 2$, $x - y = 3$ (D) $x + y = 5$, $x + y = 10$

OPTIONS: (A) $x = 0$, $x = 1$ (B) $x + y = 0$, $2x + 2y = 0$ (C) $x + y = 2$, $x - y = 3$ (D) $x + y = 5$, $x + y = 10$

ANSWER: (C) $x + y = 2$, $x - y = 3$

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The system of linear equations $px + qy = r$ and $p_1x + q_1y = r_1$ has a unique solution, if:

OPTIONS: (A) $pq \neq p_1q_1$ (B) $pp_1 \neq qq_1$ (C) $pq_1 \neq qp_1$ (D) $pqr \neq p_1q_1r_1$

ANSWER: (C) $pq_1 \neq qp_1$

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The system of equations $x = 2$ and $x = 3$ has:

OPTIONS: (A) unique solution (2, 3) (B) two solutions (2, 0) and (3, 0) (C) no solution (D) infinitely many solutions

ANSWER: (C) no solution

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If (0, 0) is the solution of the equation $x + y = c - 1$, then the value of c is:

OPTIONS: (A) 0 (B) 1 (C) -1 (D) any real number

ANSWER: (B) 1

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The system of equations $2x + ky = 4$ and $3x - 6y = 6$ has infinitely many solutions, if:

OPTIONS: (A) $k \neq 4$ (B) $k = 4$ (C) $k = -4$ (D) No value of k

ANSWER: (C) $k = -4$

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: For what value(s) of k does the system of equations $kx + 2y = 3$ and $2x + y = 5$ have a unique solution?

OPTIONS: (A) $k =$ any real number (B) $k \neq 8$ (C) $k \neq 4$ (D) $k = 4$

ANSWER: (C) $k \neq 4$

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: For what value(s) of k , is the system of equations $kx + 2y = 3$ and $2x + y = 5$ inconsistent?

OPTIONS: (A) $k =$ any real number (B) $k \neq 2$ (C) $k \neq 4$ (D) $k = 4$

ANSWER: (D) $k = 4$

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The number of solutions of the system of equations $x = a$ and $x = b$ ($a \neq b$) is:

OPTIONS: (A) 0 (B) 1 (C) 2 (D) Infinite

ANSWER: (A) 0

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The system of linear equations given by $x = a$ and $y = b$ is:

OPTIONS: (A) Consistent with a unique solution (B) Consistent with infinitely many solutions (C) Consistent with two solutions (D) Inconsistent
ANSWER: (A) Consistent with a unique solution

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Solve the following system of equations for x and y: $\frac{x}{2} + \frac{2y}{3} = -1$ and $x - \frac{y}{3} = 3$.

OPTIONS: (A) $x = 2, y = -3$ (B) $x = -2, y = 3$ (C) $x = 3, y = -2$ (D) $x = -3, y = 2$

ANSWER: (A) $x = 2, y = -3$

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find the value of c for which the following pair of linear equations has infinitely many solutions: $cx + 3y = c - 3$ and $12x + cy = c$.

OPTIONS: (A) 6 (B) -6 (C) 3 (D) -3

ANSWER: (A) 6

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Solve for x and y: $3x + 2y = 65$ and $2x + 3y = 60$.

OPTIONS: (A) $x = 15, y = 10$ (B) $x = 10, y = 15$ (C) $x = 20, y = 5$ (D) $x = 5, y = 20$

ANSWER: (A) $x = 15, y = 10$

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Solve for x and y: $0.1x + 0.3y = 1$ and $0.2x - 0.1y = -0.1$.

OPTIONS: (A) $x = 1, y = 3$ (B) $x = 3, y = 1$ (C) $x = 2, y = 2$ (D) $x = -1, y = 3$

ANSWER: (A) $x = 1, y = 3$

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Solve for x and y: $3x + 5y = 8$ and $5x - 3y = 2$.

OPTIONS: (A) $x = 1, y = 1$ (B) $x = 2, y = 0$ (C) $x = 0, y = 2$ (D) $x = 1, y = 0$

ANSWER: (A) $x = 1, y = 1$

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Solve the following system of equations graphically: $x + 3y = 6$ and $2x - 3y = 12$.

Also, find the area of the triangle formed by the lines $x + 3y = 6$, $x = 0$ and $y = 0$.

OPTIONS: (A) $x = 6, y = 0$; area = 6 (B) $x = 6, y = 0$; area = 12 (C) $x = 0, y = 6$; area = 6 (D) $x = 0, y = 6$; area = 12

ANSWER: (A) $x = 6, y = 0$; area = 6

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: x and y are complementary angles such that $x:y = 1:2$. Express the given information as a system of linear equations in two variables and hence solve it.

OPTIONS: (A) $x = 30^\circ, y = 60^\circ$ (B) $x = 60^\circ, y = 30^\circ$ (C) $x = 45^\circ, y = 45^\circ$ (D) $x = 20^\circ, y = 70^\circ$

ANSWER: (A) $x = 30^\circ, y = 60^\circ$

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that the system of equations $2x - 3y = 7$ and $4x + ky = 9$, is inconsistent for $k = -6$. Also, obtain the solution if $k = -1$.

OPTIONS: (A) inconsistent for $k = -6$, solution $x = 2, y = -1$ for $k = -1$ (B) consistent for $k = -6$, solution $x = 2, y = -1$ for $k = -1$ (C) inconsistent for $k = -6$, solution $x = -2, y = 1$ for $k = -1$ (D) consistent for $k = -6$, solution $x = -2, y = 1$ for $k = -1$

ANSWER: (A) inconsistent for $k = -6$, solution $x = 2, y = -1$ for $k = -1$

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Solve graphically the following pair of linear equations: $2x - y = 2$ and $4x - y = 4$. Also, write the coordinates of the points where the lines cut the y-axis.

OPTIONS: (A) $x = 1, y = 0$; cuts y-axis at $(0, -2)$ and $(0, -4)$ (B) $x = 0, y = 1$; cuts y-axis at $(0, -2)$ and $(0, -4)$ (C) $x = 1, y = 0$; cuts y-axis at $(0, 2)$ and $(0, 4)$ (D) $x = 0, y = 1$; cuts y-axis at $(0, 2)$ and $(0, 4)$

ANSWER: (A) $x = 1, y = 0$; cuts y-axis at $(0, -2)$ and $(0, -4)$

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: An academy bought 10 bats and 5 balls for ₹32,500. Later, bought 2 bats and 8 balls for ₹10,000. Find cost of 1 bat and 1 ball.

OPTIONS: (A) Bat = ₹3000, Ball = ₹500 (B) Bat = ₹2500, Ball = ₹1000 (C) Bat = ₹3500, Ball = ₹500 (D) Bat = ₹3000, Ball = ₹1000

ANSWER: (A) Bat = ₹3000, Ball = ₹500

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A fraction becomes $\frac{1}{3}$ when 1 is subtracted from the numerator and it becomes $\frac{1}{4}$ when 8 is added to its denominator. Find the fraction.

OPTIONS: (A) $\frac{5}{12}$ (B) $\frac{4}{13}$ (C) $\frac{7}{15}$ (D) $\frac{3}{10}$

ANSWER: (A) $\frac{5}{12}$

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the value of k for which the following pair of linear equations will have infinitely many solutions: $kx + 3y - (k - 3) = 0$ and $12x + ky - k = 0$.

OPTIONS: (A) $k = 6$ (B) $k = -6$ (C) $k = 3$ (D) $k = -3$

ANSWER: (A) $k = 6$

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The difference between two numbers is 12. The greater number is 6 less than twice the smaller one. Frame two linear equations, show unique solution, and find the numbers.

OPTIONS: (A) Numbers = 30, 18 (B) Numbers = 24, 12 (C) Numbers = 28, 16 (D) Numbers = 26, 14

ANSWER: (A) Numbers = 30, 18

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Solve the following equations graphically: $x + y = 7$ and $2x - 5y = 7$.

OPTIONS: (A) $x = 6, y = 1$ (B) $x = 5, y = 2$ (C) $x = 4, y = 3$ (D) $x = 7, y = 0$

ANSWER: (A) $x = 6, y = 1$

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A fraction becomes $\frac{5}{6}$ when 3 is added to both the numerator and the denominator. If 2 is added to both, the fraction becomes $\frac{9}{11}$. Find the original fraction.

OPTIONS: (A) $\frac{7}{9}$ (B) $\frac{5}{7}$ (C) $\frac{8}{11}$ (D) $\frac{3}{5}$

ANSWER: (A) $\frac{7}{9}$

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: Seema spent 15 min on bicycle and 30 min on walker, burned 435 calories. When she spent 30 min on bicycle and 40 min on walker, burned 690 calories. (i) Represent as linear equations. (ii) Show unique solution. (iii) Solve using elimination method.

OPTIONS: (A) Bicycle = 11 calories/min, Walker = 9 calories/min (B) Bicycle = 9 calories/min, Walker = 11 calories/min (C) Bicycle = 10 calories/min, Walker = 10 calories/min (D) Bicycle = 12 calories/min, Walker = 8 calories/min

ANSWER: (A) Bicycle = 11 calories/min, Walker = 9 calories/min

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: In Indian wedding garlands, ₹200 and ₹500 notes are used. Total notes = 40, total value = ₹11,000. Find the number of ₹200 and ₹500 notes.

OPTIONS: (A) ₹200 = 30, ₹500 = 10 (B) ₹200 = 10, ₹500 = 30 (C) ₹200 = 20, ₹500 = 20 (D) ₹200 = 25, ₹500 = 15

ANSWER: (A) ₹200 = 30, ₹500 = 10

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The sum of the ages of a father and three times the age of his daughter is 59. If the father's age is x and daughter's age is y , find the equation. (This is an MCQ, but we need to frame the equation)

OPTIONS: (A) $3x + y = 59$ (B) $x + y = 59$ (C) $x + 3y = 59$ (D) $x + y = 56$

ANSWER: (C) $x + 3y = 59$

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The sum of the age of a father and three times the age of his daughter is 59. If father's age is x years and daughter's age is y years, the equation is:

OPTIONS: (A) $3x + y = 59$ (B) $x + y = 59$ (C) $x + 3y = 59$ (D) $x + y = 56$

ANSWER: (C) $x + 3y = 59$

YEAR: 2026

CHAPTER: 3 - PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Solve for x and y : $3x + 2y = 65$ and $2x + 3y = 60$.

OPTIONS: (A) $x = 15, y = 10$ (B) $x = 10, y = 15$ (C) $x = 20, y = 5$ (D) $x = 5, y = 20$

ANSWER: (A) $x = 15, y = 10$

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CHAPTER: 4 - QUADRATIC EQUATIONS

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YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The discriminant of the quadratic equation $x^2 - 3x - 2 = 0$ is:

OPTIONS: (A) 1 (B) 17 (C) $\sqrt{17}$ (D) $-\sqrt{17}$

ANSWER: (B) 17

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The discriminant of the quadratic equation $ax^2 + x + a = 0$ is:

OPTIONS: (A) $\sqrt{1 - 4a^2}$ (B) $1 - 4a^2$ (C) $4a^2 - 1$ (D) $\sqrt{4a^2 - 1}$

ANSWER: (B) $1 - 4a^2$

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The roots of the quadratic equation $x^2 + 9 = 0$ are:

OPTIONS: (A) real and equal (B) not real (C) real and negative of each other (D) rational numbers

ANSWER: (B) not real

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of k for which the equation $2kx^2 - 6x + 3 = 0$ has real and equal roots is:

OPTIONS: (A) $3/2$ (B) $1/2$ (C) $-3/2$ (D) 2

ANSWER: (A) $3/2$

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The discriminant of the quadratic equation $-x^2 - 5x + 6 = 0$ is:

OPTIONS: (A) 1 (B) -1 (C) 49 (D) 7

ANSWER: (C) 49

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The equation $x + 1/x = 3$ ($x \neq 0$) is expressed as a quadratic equation in the form $ax^2 + bx + c = 0$. The value of $a - b + c$ is:

OPTIONS: (A) 5 (B) 2 (C) 1 (D) -1

ANSWER: (C) 1

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of 'a' for which $ax^2 + 3x + 1 = 0$ has real and equal roots is:

OPTIONS: (A) $4/9$ (B) $9/4$ (C) $3/2$ (D) $2/3$

ANSWER: (B) $9/4$

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following equations is a quadratic equation? (A) $x^2 = (x+1)^2$ (B) $(x-1)(x+2) = 2x+1$ (C) $(x+2)^3 = 2x(x^2-1)$ (D) $\sqrt{x} = x^2$

OPTIONS: (A) $x^2 = (x+1)^2$ (B) $(x-1)(x+2) = 2x+1$ (C) $(x+2)^3 = 2x(x^2-1)$ (D) $\sqrt{x} = x^2$

ANSWER: (B) $(x-1)(x+2) = 2x+1$

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If the quadratic equation $ax^2 + 2bx - c = 0$ ($a \neq 0$) has real and equal roots, then which of the following is true?

OPTIONS: (A) $b^2 = -ac$ (B) $b^2 = 4ac$ (C) $b^2 = ac$ (D) $b^2 = -4ac$

ANSWER: (A) $b^2 = -ac$

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $x = -1$ is a root of the equation $ax^2 - bx + 3 = 0$, then:

OPTIONS: (A) $-a + b - 3 = 0$ (B) $a - b - 3 = 0$ (C) $-a - b + 3 = 0$ (D) $a + b + 3 = 0$

ANSWER: (D) $a + b + 3 = 0$

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of 'a' for which the equation $x^2 + 3x + a = 0$ has real and equal roots is:

OPTIONS: (A) $4/9$ (B) $9/4$ (C) $3/2$ (D) $2/3$

ANSWER: (B) $9/4$

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If the quadratic equation $lx^2 - mx + n = 0$ has roots which are reciprocal of each other, then which is true?

OPTIONS: (A) $l = n$ (B) $l = m$ (C) $m = n$ (D) $l = 1/n$

ANSWER: (A) $l = n$

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The discriminant of the quadratic equation $2x^2 - 3x - 5 = 0$ is:

OPTIONS: (A) -31 (B) 49 (C) 7 (D) $\sqrt{-31}$

ANSWER: (B) 49

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of 'm' for which the equation $x^2 - mx - m = 0$ has real and equal roots is:

OPTIONS: (A) 0 only (B) 4 only (C) 0 or 4 (D) 0 or -4

ANSWER: (D) 0 or -4

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of 'p' for which the equation $x^2 - 2x - p = 0$ has real and equal roots is:

OPTIONS: (A) -1 (B) 0 (C) 1 (D) 4

ANSWER: (A) -1

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of k for which the equation $5x^2 - 2x + k = 0$ has equal real roots is:

OPTIONS: (A) 5 (B) $-1/5$ (C) $1/5$ (D) 0

ANSWER: (C) $1/5$

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $\sqrt{x} = x^2$ ($x \neq 0$) is expressed as a quadratic equation $ax^2 + bx + c = 0$, then the value of $a + b + c$ is:

OPTIONS: (A) 0 (B) 1 (C) 2 (D) 3

ANSWER: (A) 0

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $(x + 1)^3 = x^3 + 1$ is expressed as a quadratic equation $px^2 + qx + r = 0$, then the value of $p - q + r$ is:

OPTIONS: (A) 0 (B) 1 (C) 3 (D) 6

ANSWER: (C) 3

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: (a) Express $24/(18 - x) - 24/(18 + x) = 1$ as a quadratic equation and find the discriminant and roots.

OPTIONS: (A) $x^2 + 48x - 324 = 0$; $D = 3600$; roots = 6, -54 (B) $x^2 - 48x - 324 = 0$; $D = 3600$; roots = 6, -54 (C) $x^2 + 48x - 324 = 0$; $D = 3600$; roots = -6, 54 (D) $x^2 - 48x - 324 = 0$; $D = 3600$; roots = -6, 54

ANSWER: (A) $x^2 + 48x - 324 = 0$; $D = 3600$; roots = 6, -54

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: (b) The sum of squares of two positive numbers is 100. If one number exceeds the other by 2, find the numbers.

OPTIONS: (A) 6 and 8 (B) 5 and 7 (C) 4 and 6 (D) 8 and 10

ANSWER: (A) 6 and 8

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Find the value(s) of k for which the equation $2x^2 + kx + 3 = 0$ has real and equal roots. Hence, find the roots.

OPTIONS: (A) $k = \pm 2\sqrt{6}$; roots = $-\sqrt{6}/2, -\sqrt{6}/2$ (B) $k = \pm 2\sqrt{6}$; roots = $\sqrt{6}/2, \sqrt{6}/2$ (C) $k = \pm\sqrt{6}$; roots = $-\sqrt{6}, -\sqrt{6}$ (D) $k = \pm\sqrt{6}$; roots = $\sqrt{6}, \sqrt{6}$

ANSWER: (A) $k = \pm 2\sqrt{6}$; roots = $-\sqrt{6}/2, -\sqrt{6}/2$

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Express $1/x - 1/(x-2) = 3$ ($x \neq 0, 2$) as a quadratic equation in standard form and find its roots.

OPTIONS: (A) $3x^2 - 6x + 2 = 0$; roots = $(3 \pm \sqrt{3})/3$ (B) $3x^2 - 6x - 2 = 0$; roots = $(3 \pm \sqrt{3})/3$ (C) $3x^2 - 6x + 2 = 0$; roots = $(3 \pm \sqrt{3})/2$ (D) $3x^2 - 6x - 2 = 0$; roots = $(3 \pm \sqrt{3})/2$

ANSWER: (A) $3x^2 - 6x + 2 = 0$; roots = $(3 \pm \sqrt{3})/3$

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A train covers a distance of 90 km at a uniform speed. Had the speed been 15 km/h more, it would have taken 30 minutes less. Find the original speed.

OPTIONS: (A) 45 km/h (B) 60 km/h (C) 30 km/h (D) 75 km/h

ANSWER: (A) 45 km/h

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The sum of areas of two squares is 2650 cm^2 . If the sum of their perimeters is 280 cm, find the sides of the squares.

OPTIONS: (A) 25 cm and 45 cm (B) 30 cm and 40 cm (C) 20 cm and 50 cm (D) 15 cm and 55 cm

ANSWER: (A) 25 cm and 45 cm

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Find two consecutive odd integers, sum of whose squares is 290.

OPTIONS: (A) 11 and 13 (B) 13 and 15 (C) 9 and 11 (D) 15 and 17

ANSWER: (A) 11 and 13

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A charity trust decides to build a rectangular hall with area 300 m^2 . The length is one metre more than twice its width. Find length and breadth.

OPTIONS: (A) Length = 25 m, Breadth = 12 m (B) Length = 20 m, Breadth = 15 m (C) Length = 30 m, Breadth = 10 m (D) Length = 24 m, Breadth = 12.5 m

ANSWER: (A) Length = 25 m, Breadth = 12 m

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Express $x - \frac{1}{x} = 3$ as a quadratic equation and find its roots. Also, find 'a' such that $x + \frac{1}{x} = a$ has real and equal roots.

OPTIONS: (A) $x^2 - 3x - 1 = 0$; roots = $(3 \pm \sqrt{13})/2$; $a = \pm 2$ (B) $x^2 - 3x + 1 = 0$; roots = $(3 \pm \sqrt{5})/2$; $a = \pm 2$ (C) $x^2 - 3x - 1 = 0$; roots = $(3 \pm \sqrt{13})/2$; $a = \pm 1$ (D) $x^2 - 3x + 1 = 0$; roots = $(3 \pm \sqrt{5})/2$; $a = \pm 1$
ANSWER: (A) $x^2 - 3x - 1 = 0$; roots = $(3 \pm \sqrt{13})/2$; $a = \pm 2$

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The difference of the squares of two positive numbers is 180. The square of the smaller number is 8 times the greater number. Find the numbers.

OPTIONS: (A) 18 and 12 (B) 20 and 10 (C) 24 and 12 (D) 16 and 8

ANSWER: (A) 18 and 12

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: The fencing problem: Six identical rectangular enclosures, each x feet by y feet, total fencing length 152 feet and each area 80 sq ft. (i) Write expression for fencing length, (ii) area in terms of x , (iii) solve quadratic equation by factorisation to find dimensions.

OPTIONS: (A) $x = 10$, $y = 8$ (B) $x = 9$, $y = 10$ (C) $x = 8$, $y = 10$ (D) $x = 10$, $y = 9$

ANSWER: (A) $x = 10$, $y = 8$

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: The fencing problem: Six identical rectangular enclosures, each x feet by y feet, total fencing length 152 feet and each area 80 sq ft. (i) Write expression for fencing length, (ii) area in terms of x , (iii) solve quadratic equation by quadratic formula.

OPTIONS: (A) $x = 10$, $y = 8$ (B) $x = 9$, $y = 10$ (C) $x = 8$, $y = 10$ (D) $x = 10$, $y = 9$

ANSWER: (A) $x = 10$, $y = 8$

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: ABCD is a rectangle of dimensions 80 cm $\times$ 60 cm. Another rectangle PQRS is drawn inside leaving equal width x cm. If area PQRS is half of area ABCD, find x .

OPTIONS: (A) 10 cm (B) 15 cm (C) 5 cm (D) 20 cm

ANSWER: (A) 10 cm

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A quadratic equation $ax^2 + bx + c = 0$ has roots which are 1 less than the roots of $5x^2 - 2x - 3 = 0$. Find the roots of the equation $ax^2 + bx + c = 0$.

OPTIONS: (A) $-2/3$ and 2 (B) $-4/3$ and 0 (C) $5/3$ and 3 (D) $2/3$ and -2

ANSWER: (C) $5/3$ and 3

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A quadratic equation $ax^2 + bx + c = 0$ has roots which are twice the roots of $2x^2 - 5x + 2 = 0$. Find the roots of the equation $ax^2 + bx + c = 0$.

OPTIONS: (A) 4 and 1 (B) 2 and 1 (C) 4 and 2 (D) 8 and 2

ANSWER: (A) 4 and 1

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A fraction becomes $1/3$ when 1 is subtracted from numerator and $1/4$ when 8 is added to denominator. Find the fraction. (This is also solved via linear equations, but can be framed as quadratic if cross-multiplied; but it's linear. I'll keep it here as it's from the syllabus.)

OPTIONS: (A) $5/12$ (B) $4/13$ (C) $7/15$ (D) $3/10$

ANSWER: (A) $5/12$

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: If $\sin A + \sin^2 A = 1$, find $\cos^2 A + \cos^4 A$. Also prove that $\tan^2 A \cdot \sec^2 A = 1$. (This is trigonometric but involves quadratic relation; I'll include it here as it uses quadratic manipulation.)

OPTIONS: (A) $\cos^2 A + \cos^4 A = 1$ (B) $\cos^2 A + \cos^4 A = 0$ (C) $\cos^2 A + \cos^4 A = 2$ (D) $\cos^2 A + \cos^4 A = 1/2$

ANSWER: (A) $\cos^2 A + \cos^4 A = 1$

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following equations is/are quadratic equation(s)? q1: $x^2 + x = (x+1)^2$, q2: $x - 1 = x^2 - 1$, q3: $x^4 = x^2$, q4: $\sqrt{x} = x^2\sqrt{x} + 1$.

OPTIONS: (A) q1 only (B) q1, q2 and q3 only (C) q2 only (D) q2 and q4 only

ANSWER: (C) q2 only

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following equations is/are not quadratic? q1: $(x-1)^2 = x^2$, q2: $(x-1)^3 = x^3$, q3: $(x+1)^3 = 3x^2$, q4: $(\sqrt{x} + 1)^2 = 2\sqrt{x}$.

OPTIONS: (A) q1, q3 and q4 (B) q2 and q3 (C) q2, q3 and q4 (D) q1 and q4

ANSWER: (A) q1, q3 and q4

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $(\sqrt{x} + 1)^2 = x^2 + 2\sqrt{x}$ is expressed as a quadratic equation in the form $ax^2 + bx + c = 0$, then the value of $a - b + c$ is:

OPTIONS: (A) -1 (B) 0 (C) 1 (D) 2

ANSWER: (B) 0

YEAR: 2026

CHAPTER: 4 - QUADRATIC EQUATIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following statements is true for the quadratic equation $ax^2 + x + a = 0$ ($a \neq 0$)?

OPTIONS: (A) The roots are reciprocal of each other (B) The roots are always real (C) The roots are always positive (D) The sum of roots is 0

ANSWER: (A) The roots are reciprocal of each other

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CHAPTER: 5 - ARITHMETIC PROGRESSIONS

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YEAR: 2026
CHAPTER: 5 - ARITHMETIC PROGRESSIONS
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The 10th term of the A.P. $\sqrt{2}, \sqrt{8}, \sqrt{18}, \dots$ is:
OPTIONS: (A) $\sqrt{162}$ (B) $\sqrt{200}$ (C) $\sqrt{54}$ (D) $\sqrt{94}$
ANSWER: (B) $\sqrt{200}$

YEAR: 2026
CHAPTER: 5 - ARITHMETIC PROGRESSIONS
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The numbers $x, x+4$ and $x+8$ are in A.P. with common difference:
OPTIONS: (A) x (B) $4+x$ (C) 4 (D) 0
ANSWER: (C) 4

YEAR: 2026
CHAPTER: 5 - ARITHMETIC PROGRESSIONS
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: If the numbers $2p+1, 3p+2, 4p+3$ are in A.P., then the common difference is:
OPTIONS: (A) p (B) 1 (C) $p+1$ (D) 0
ANSWER: (C) $p+1$

YEAR: 2026
CHAPTER: 5 - ARITHMETIC PROGRESSIONS
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: If the numbers $1-a, 3-a$ and $5-a$ are in A.P., then the common difference is:
OPTIONS: (A) $1-a$ (B) $-2+a$ (C) 2 (D) $2-a$
ANSWER: (C) 2

YEAR: 2026
CHAPTER: 5 - ARITHMETIC PROGRESSIONS
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1

QUESTION: If a , b and c are in A.P., then:

OPTIONS: (A) $2a$, $3b$ and $4c$ are also in A.P. (B) $a-2$, $b-2$ and $c-2$ are also in A.P. (C) $a/2$, $b/3$, $c/4$ are also in A.P. (D) $a+2$, $b+3$, $c+4$ are also in A.P.

ANSWER: (B) $a-2$, $b-2$ and $c-2$ are also in A.P.

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $p-1$, $q-1$ and $r-1$ are in A.P., then:

OPTIONS: (A) $p-2$, q , $r-2$ are also in A.P. (B) $2p$, $3q$, $4r$ are also in A.P. (C) $p/2$, $q/3$, $r/4$ are also in A.P. (D) $2p$, $2q$, $2r$ are also in A.P.

ANSWER: (D) $2p$, $2q$ and $2r$ are also in A.P.

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $x+1$, $y+1$ and $z+1$ are in A.P., then which of the following are in A.P.?

OPTIONS: (A) $2x$, $3y$, $4z$ (B) $x+1$, $2y+1$, $3z+1$ (C) $x-2$, $y-3$, $z-4$ (D) $x-3$, $y-3$, $z-3$

ANSWER: (D) $x-3$, $y-3$, $z-3$

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: n th term of the A.P.: $-1/3$, $4/3$, 3 , ... is:

OPTIONS: (A) $(5n-9)/3$ (B) $(5n-6)/3$ (C) $(3n-4)/3$ (D) $(3n+2)/3$

ANSWER: (B) $(5n-6)/3$

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: n th term of the A.P.: $-3/2$, $3/2$, $9/2$, ... is:

OPTIONS: (A) $3n/2 - 3$ (B) $3n - 9/2$ (C) $(3n-9)/2$ (D) $3n + 3/2$

ANSWER: (B) $3n - 9/2$

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: n th term of an A.P. is $5n - 15$. The common difference of the A.P. is:

OPTIONS: (A) $5n$ (B) 5 (C) -5 (D) 10

ANSWER: (B) 5

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If -26, x , 2 are in A.P., then the value of x is:

OPTIONS: (A) 14 (B) -13 (C) -12 (D) -14

ANSWER: (C) -12

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $-20/9$, $-2/9$, $16/9$, ... are in A.P., then the next term is:

OPTIONS: (A) $32/9$ (B) $46/9$ (C) $2/9$ (D) $34/9$

ANSWER: (D) $34/9$

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If the 14th term of an A.P. is 4 and its 15th term is zero, then its first term is:

OPTIONS: (A) -48 (B) -56 (C) 56 (D) 48

ANSWER: (C) 56

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: The sum of first n terms of an A.P. is $50\sqrt{2}$. If the first and last terms are $\sqrt{2}$ and $19\sqrt{2}$ respectively, find n .

OPTIONS: (A) 10 (B) 5 (C) 15 (D) 20

ANSWER: (B) 5

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: In an A.P., it is given that $a = 2$, $d = 8$ and $S_n = 90$. Find the value of n .

OPTIONS: (A) 3 (B) 5 (C) 7 (D) 9

ANSWER: (B) 5

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: How many 4-digit numbers are divisible by 7?

OPTIONS: (A) 1286 (B) 1285 (C) 1284 (D) 1287

ANSWER: (A) 1286

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: How many terms of the A.P. 3, 5, 7, 9, ... must be added to get the sum 80?

OPTIONS: (A) 6 (B) 8 (C) 10 (D) 12

ANSWER: (B) 8

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find three consecutive terms in A.P. whose sum is 21 and their product is 231.

OPTIONS: (A) 3, 7, 11 (B) 5, 7, 9 (C) 1, 7, 13 (D) 3, 5, 13

ANSWER: (A) 3, 7, 11

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: The 4th and 10th term of an A.P. are 13 and 25 respectively. Find its 24th term.

OPTIONS: (A) 50 (B) 52 (C) 53 (D) 55

ANSWER: (C) 53

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: From 232 to 540, find the number of multiples of 3.

OPTIONS: (A) 100 (B) 101 (C) 102 (D) 103

ANSWER: (D) 103

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The third and ninth term of an A.P. are 4 and -8 respectively. (i) Which term is zero? (ii) Find n if $S_n = -36$.

OPTIONS: (A) 5th term is zero, $n = 12$ (B) 6th term is zero, $n = 10$ (C) 4th term is zero, $n = 15$ (D) 5th term is zero, $n = 10$

ANSWER: (A) 5th term is zero, $n = 12$

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A multistorey building has stairs from ground to first floor = 10, first to second = 24, second to third = 38, and so on. (i) Does 10, 24, 38 form an A.P.? (ii) Find total stairs from ground to eleventh floor. (iii)(a) Find total stairs climbed up and down to supply water till sixth floor.

OPTIONS: (A) Yes, $d=14$; 880; 1400 (B) No; 880; 1400 (C) Yes, $d=14$; 770; 1400 (D) Yes, $d=14$; 880; 1200

ANSWER: (A) Yes, $d=14$; 880; 1400

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A multistorey building has stairs from ground to first floor = 10, first to second = 24, second to third = 38, and so on. (i) Does 10, 24, 38 form an A.P.? (ii) Find total stairs from ground to eleventh floor. (iii)(b) If a person climbed up and down a total of 380 stairs, till which floor did he supply water?

OPTIONS: (A) 3rd floor (B) 4th floor (C) 5th floor (D) 6th floor

ANSWER: (A) 3rd floor

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A woman borrowed ₹10,00,000 and returned ₹10,000 in first month, ₹15,000 in second, ₹20,000 in third, and so on. (i) Find amount paid in tenth month. (ii) In which instalment did she pay ₹40,000? (iii)(a) If she returned ₹11,50,000 in all, how many instalments did she pay?

OPTIONS: (A) 10th month = ₹55,000; 7th instalment; 20 instalments (B) 10th month = ₹55,000; 7th instalment; 15 instalments (C) 10th month = ₹50,000; 8th instalment; 20 instalments (D) 10th month = ₹55,000; 8th instalment; 15 instalments

ANSWER: (A) 10th month = ₹55,000; 7th instalment; 20 instalments

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A woman borrowed ₹10,00,000 and returned ₹10,000 in first month, ₹15,000 in second, ₹20,000 in third, and so on. (i) Find amount paid in tenth month. (ii) In which instalment did she pay ₹40,000? (iii)(b) By which instalment has she returned a total of ₹3,25,000?

OPTIONS: (A) 10th instalment (B) 12th instalment (C) 15th instalment (D) 20th instalment

ANSWER: (A) 10th instalment

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A watermelon vendor arranged watermelons in rows. Bottommost row has 101 watermelons and topmost row has 1 watermelon. There are 21 rows. (i) Find the common difference. (ii) Find number of watermelons in 15th row. (iii)(a) Find total watermelons in the arrangement.

OPTIONS: (A) $d = -5$; 15th row = 31; total = 1071 (B) $d = -5$; 15th row = 31; total = 1071 (C) $d = -5$; 15th row = 26; total = 1071 (D) $d = -5$; 15th row = 31; total = 1021

ANSWER: (A) $d = -5$; 15th row = 31; total = 1071

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A watermelon vendor arranged watermelons in rows. Bottommost row has 101 watermelons and topmost row has 1 watermelon. There are 21 rows. (i) Find the common difference. (ii) Find number of watermelons in 15th row. (iii)(b) Which row has 1 watermelon?

OPTIONS: (A) 21st row (B) 20th row (C) 19th row (D) 18th row

ANSWER: (A) 21st row

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A city based NGO built a human pyramid with 11 levels. Level 1 (bottom) has 41 members, level 2 has 37, level 3 has 33, and so on. (i) Find members at level eleven. (ii) Find members at third level from top. (iii)(a) Find total members who formed the pyramid.

OPTIONS: (A) 1; 9; 231 (B) 1; 9; 231 (C) 1; 13; 231 (D) 1; 9; 241

ANSWER: (A) 1; 9; 231

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A city based NGO built a human pyramid with 11 levels. Level 1 (bottom) has 41 members, level 2 has 37, level 3 has 33, and so on. (i) Find members at level eleven. (ii) Find members at third level from top. (iii)(b) At which level is the number of members 5 times the number of members at level ten?

OPTIONS: (A) 5th level (B) 6th level (C) 7th level (D) 8th level

ANSWER: (A) 5th level

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A spiral pattern of semicircles has radii 50 cm, 100 cm, 150 cm, ... Spiral 1 has 10 flowers, Spiral 2 has 20 flowers, Spiral 3 has 30 flowers, and so on. (i) Find radius of 13th spiral. (ii) If radius of nth spiral is 500 cm, find n. (iii)(a) Find total saplings till 11th spiral.

OPTIONS: (A) 650 cm; n=10; 660 saplings (B) 650 cm; n=10; 660 saplings (C) 600 cm; n=10; 660 saplings (D) 650 cm; n=11; 660 saplings

ANSWER: (A) 650 cm; n=10; 660 saplings

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A spiral pattern of semicircles has radii 50 cm, 100 cm, 150 cm, ... Spiral 1 has 10 flowers, Spiral 2 has 20 flowers, Spiral 3 has 30 flowers, and so on. (i) Find radius of 13th spiral. (ii) If radius of nth spiral is 500 cm, find n. (iii)(b) Till which spiral will there be a total of 450 saplings?

OPTIONS: (A) 9th spiral (B) 10th spiral (C) 8th spiral (D) 11th spiral

ANSWER: (A) 9th spiral

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A yoga instructor was hired. On day one, 5 people joined, on day two, 3 more people joined, on day three, another 3 more joined, and so on. (i) On which day did 59 people join? (ii) How many joined on the 31st day? (iii)(a) If paid ₹100 per person, on which day would he earn ₹5,000?

OPTIONS: (A) 19th day; 95; 17th day (B) 19th day; 95; 17th day (C) 18th day; 95; 17th day (D) 19th day; 90; 17th day

ANSWER: (A) 19th day; 95; 17th day

YEAR: 2026

CHAPTER: 5 - ARITHMETIC PROGRESSIONS

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A yoga instructor was hired. On day one, 5 people joined, on day two, 3 more people joined, on day three, another 3 more joined, and so on. (i) On which day did 59 people join? (ii) How many joined on the 31st day? (iii)(b) What was the total amount earned in 16 days?

OPTIONS: (A) ₹12,800 (B) ₹13,600 (C) ₹12,000 (D) ₹14,400

ANSWER: (A) ₹12,800

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CHAPTER: 6 - COORDINATE GEOMETRY

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YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The co-ordinates of the mid-point of the line segment joining points A(p-1, q+1) and B(p+1, q-1) are given by:

OPTIONS: (A) (p, q) (B) (2p, 2q) (C) $((p+1)/2, (q+1)/2)$ (D) $((p-1)/2, (q-1)/2)$

ANSWER: (A) (p, q)

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The distance between the points (3, 0) and (0, -3) is:

OPTIONS: (A) 3 units (B) 6 units (C) $\sqrt{6}$ units (D) $\sqrt{18}$ units

ANSWER: (D) $\sqrt{18}$ units

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The distance between the points (-2, 5) and (5, -2) is:

OPTIONS: (A) $7\sqrt{2}$ (B) 14 (C) $2\sqrt{7}$ (D) 7

ANSWER: (A) $7\sqrt{2}$

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The distance between the points (-4, 5) and (-1, 2) is:

OPTIONS: (A) 5 (B) $3\sqrt{2}$ (C) 6 (D) $2\sqrt{3}$

ANSWER: (B) $3\sqrt{2}$

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The distance between the points $(-5, 1)$ and $(2, 2)$ is:

OPTIONS: (A) $2\sqrt{5}$ units (B) $\sqrt{10}$ units (C) $5\sqrt{2}$ units (D) $3\sqrt{2}$ units

ANSWER: (C) $5\sqrt{2}$ units

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The distance between the points $(-4, 2)$ and $(1, 0)$ is:

OPTIONS: (A) $\sqrt{13}$ units (B) 3 units (C) 9 units (D) $\sqrt{29}$ units

ANSWER: (D) $\sqrt{29}$ units

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The distance between the points $(-1, 2\sqrt{2})$ and $(2, \sqrt{2})$ is:

OPTIONS: (A) $\sqrt{5}$ units (B) $\sqrt{11}$ units (C) $\sqrt{13}$ units (D) $\sqrt{7}$ units

ANSWER: (B) $\sqrt{11}$ units

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The end points of the diameter AB of a circle are $A(4, 0)$ and $B(0, -4)$. The length of the diameter is:

OPTIONS: (A) 3 units (B) 8 units (C) $\sqrt{8}$ units (D) $\sqrt{32}$ units

ANSWER: (D) $\sqrt{32}$ units

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The coordinates of opposite vertices of the square ABCD are $A(-5, 0)$ and $C(0, 5)$. The length of a diagonal of the square ABCD is:

OPTIONS: (A) 5 units (B) 10 units (C) $\sqrt{10}$ units (D) $\sqrt{50}$ units

ANSWER: (D) $\sqrt{50}$ units

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The distance of point $(-3, 4)$ from y-axis is:

OPTIONS: (A) -3 (B) 3 (C) 4 (D) 5

ANSWER: (B) 3

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The distance of a point $P(3, -7)$ from y-axis is:

OPTIONS: (A) 3 (B) 7 (C) -7 (D) $\sqrt{58}$

ANSWER: (A) 3

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: For a point $(3, -5)$, the value of (abscissa - ordinate) is:

OPTIONS: (A) -8 (B) -2 (C) 2 (D) 8

ANSWER: (D) 8

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: For a point $X(a, b)$ where $b > a > 0$, the value of its [distance from x-axis - distance from y-axis] is:

OPTIONS: (A) $a - b$ (B) $b - a$ (C) $a^2 - b^2$ (D) $b^2 - a^2$

ANSWER: (B) $b - a$

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The mid-point of the line segment joining points $(1, 3)$ and $(1, -3)$ lies:

OPTIONS: (A) at the origin (B) in the second quadrant (C) on x-axis (D) on y-axis

ANSWER: (D) on y-axis

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The mid-point of a line segment divides the line segment in the ratio:

OPTIONS: (A) 1:2 (B) 2:1 (C) 1:1 (D) 1/2:2

ANSWER: (C) 1:1

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find the coordinates of the point which divides the line segment joining the points A(-6, 10) and B(3, -8) in the ratio 2:7.

OPTIONS: (A) (-4, 6) (B) (4, -6) (C) (-2, 4) (D) (2, -4)

ANSWER: (A) (-4, 6)

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find the coordinates of the point which divides the line segment joining the points P(-1, 1) and Q(5, -7) in the ratio 2:3.

OPTIONS: (A) (7/5, -11/5) (B) (-7/5, 11/5) (C) (11/5, -7/5) (D) (-11/5, 7/5)

ANSWER: (A) (7/5, -11/5)

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: A point P divides a line segment OR in the ratio 2:1. If the coordinates of O are (0, 0) and that of P are (-2, -3), find the coordinates of R.

OPTIONS: (A) (-3, -9/2) (B) (-3, -9/2) (C) (-3, -9/2) (D) (-3, -9/2)

ANSWER: (A) (-3, -9/2)

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If point $(1, 2)$ divides the line segment joining the points $(3, 5)$ and $(2p, q)$ in the ratio $1:1$, then (p, q) is equal to:

OPTIONS: (A) $(-1/2, -1)$ (B) $(-1/2, -1/2)$ (C) $(-1, -1)$ (D) $(-1, -1/2)$

ANSWER: (A) $(-1/2, -1)$

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If point $(1, 2)$ is the mid-point of the line segment joining the points $(3, 5)$ and $(2a, b)$, then (a, b) is:

OPTIONS: (A) $(-1, -1)$ (B) $(-1/2, -1/2)$ (C) $(-1/2, -1)$ (D) $(-1, -1/2)$

ANSWER: (C) $(-1/2, -1)$

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If $A(a, 0)$, $B(1, 1)$ and $C(0, b)$ form a triangle, right angled at B when joined, then establish a relation between a and b .

OPTIONS: (A) $a + b = 2$ (B) $a + b = 1$ (C) $a - b = 2$ (D) $ab = 1$

ANSWER: (A) $a + b = 2$

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: $A(-2, 3)$ and $B(4, 1)$ are end points of the diameter of a semi-circle. The semi-circle intersects y -axis at point P . Find the co-ordinates of P .

OPTIONS: (A) $(0, 5)$ or $(0, -1)$ (B) $(0, 5)$ only (C) $(0, -1)$ only (D) $(0, 3)$ or $(0, 1)$

ANSWER: (A) $(0, 5)$ or $(0, -1)$

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Show that the quadrilateral $ABCD$ with vertices $A(0, 3)$, $B(-2, 0)$, $C(0, -5)$ and $D(2, 0)$ is a kite. Also, find the length of each diagonal.

OPTIONS: (A) Diagonals: $AC = 8$, $BD = 4$ (B) Diagonals: $AC = 10$, $BD = 4$ (C) Diagonals: $AC = 8$, $BD = 6$ (D) Diagonals: $AC = 6$, $BD = 4$
ANSWER: (A) Diagonals: $AC = 8$, $BD = 4$

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CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Show that the quadrilateral 'HOPE' with vertices $H(-2,1)$, $O(-1,2)$, $P(0,1)$ and $E(-1,0)$ is a rhombus. Is it a square?

OPTIONS: (A) Rhombus, yes square (B) Rhombus, not square (C) Not a rhombus, not square (D) Not a rhombus, yes square

ANSWER: (A) Rhombus, yes square

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Show that the quadrilateral PLOT with vertices $P(-1, 0)$, $L(2, 6)$, $O(5, 0)$, $T(2, -6)$ is a rectangle. Is it a square?

OPTIONS: (A) Rectangle, not square (B) Rectangle, yes square (C) Not a rectangle, not square (D) Not a rectangle, yes square

ANSWER: (A) Rectangle, not square

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Show that the quadrilateral ABCD formed by joining the points $A(-2, -1)$, $B(1, -1)$, $C(1, 2)$ and $D(-2, 2)$ in order is a square.

OPTIONS: (A) Yes, square (B) No, rectangle only (C) No, rhombus only (D) No, parallelogram only

ANSWER: (A) Yes, square

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CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Find the coordinates of the points which divide the line segment joining points $P(-1, 7)$ and $Q(4, -3)$ into three equal parts.

OPTIONS: (A) $(\frac{2}{3}, \frac{11}{3})$ and $(\frac{7}{3}, \frac{1}{3})$ (B) $(\frac{2}{3}, \frac{11}{3})$ and $(\frac{7}{3}, \frac{1}{3})$ (C) $(\frac{1}{3}, \frac{11}{3})$ and $(\frac{7}{3}, \frac{1}{3})$ (D) $(\frac{2}{3}, \frac{11}{3})$ and $(\frac{5}{3}, \frac{1}{3})$

ANSWER: (A) $(\frac{2}{3}, \frac{11}{3})$ and $(\frac{7}{3}, \frac{1}{3})$

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: The line segment joining the points A(-5, 1) and B(7, 6) is trisected at points P and Q such that P is nearer to A. If P lies on the line $x + y = k$, then find k.

OPTIONS: (A) $\frac{5}{3}$ (B) $\frac{3}{5}$ (C) $\frac{7}{3}$ (D) $\frac{3}{7}$

ANSWER: (A) $\frac{5}{3}$

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: The vertices of a rhombus ABCD are A(-3, -4), B(5, -3), C(1, 4) and D(-7, 3). Find the length of both the diagonals and the area of the rhombus.

OPTIONS: (A) Diagonals = $4\sqrt{5}$, $6\sqrt{5}$; Area = 60 (B) Diagonals = $4\sqrt{5}$, $6\sqrt{5}$; Area = 120 (C) Diagonals = $4\sqrt{5}$, $6\sqrt{5}$; Area = 30 (D) Diagonals = $4\sqrt{5}$, $6\sqrt{5}$; Area = 60

ANSWER: (A) Diagonals = $4\sqrt{5}$, $6\sqrt{5}$; Area = 60

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A circular park has two gates at A(10, 20) and B(50, 50). Two fountains P and Q on AB such that AP = PQ = QB. (i) Find coordinates of centre C. (ii) Find radius. (iii)(a) Find coordinates of P.

OPTIONS: (A) C = (30, 35); Radius = 25; P = $(\frac{70}{3}, 30)$ (B) C = (30, 35); Radius = 25; P = $(\frac{70}{3}, 30)$ (C) C = (30, 35); Radius = 30; P = $(\frac{70}{3}, 30)$ (D) C = (30, 35); Radius = 25; P = $(\frac{80}{3}, 30)$

ANSWER: (A) C = (30, 35); Radius = 25; P = $(\frac{70}{3}, 30)$

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A circular park has two gates at A(10, 20) and B(50, 50). Two fountains P and Q on AB such that $AP = PQ = QB$. (i) Find coordinates of centre C. (ii) Find radius. (iii)(b) Find distance of fountain at Q from gate A.

OPTIONS: (A) $100/3$ m (B) $50/3$ m (C) $100/5$ m (D) $200/3$ m

ANSWER: (A) $100/3$ m

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CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A rectangular field ABCD has coordinates A(10, 10), B(40, 10), C(40, 50) and D(x, y). Anita ran from A to E (intersection of diagonals), then to B, then back to A. Anil ran from C to A via D. (i) Find coordinates of E. (ii) Find distance BC. (iii)(a) Find coordinates of D and distance BD.

OPTIONS: (A) E = (25, 30); BC = 40; D = (10, 50); BD = 50 (B) E = (25, 30); BC = 40; D = (10, 50); BD = 50 (C) E = (25, 30); BC = 40; D = (10, 50); BD = 40 (D) E = (25, 30); BC = 50; D = (10, 50); BD = 50

ANSWER: (A) E = (25, 30); BC = 40; D = (10, 50); BD = 50

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A rectangular field ABCD has coordinates A(10, 10), B(40, 10), C(40, 50) and D(x, y). Anita ran from A to E (intersection of diagonals), then to B, then back to A. Anil ran from C to A via D. (i) Find coordinates of E. (ii) Find distance BC. (iii)(b) Find total distance travelled by Anita.

OPTIONS: (A) $40 + 50 + 30 = 120$ (B) $40 + 50 + 30 = 120$ (C) $40 + 50 + 20 = 110$ (D) $30 + 50 + 30 = 110$

ANSWER: (A) $40 + 50 + 30 = 120$

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: There is a semicircular park with diameter AB. A borewell at centre O. (i) Find coordinates of point O. (ii) Find radius of the semicircular park. (iii)(a) One sapling is kept at point C(12, y). Find the coordinates of C.

OPTIONS: (A) O = (12, 0); Radius = 13; C = (12, 5) (B) O = (12, 0); Radius = 13; C = (12, 5) (C) O = (12, 0); Radius = 13; C = (12, 5) (D) O = (12, 0); Radius = 13; C = (12, 5)

ANSWER: (A) O = (12, 0); Radius = 13; C = (12, 5)

YEAR: 2026

CHAPTER: 6 - COORDINATE GEOMETRY

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: There is a semicircular park with diameter AB. A borewell at centre O. (i) Find coordinates of point O. (ii) Find radius of the semicircular park. (iii)(b) One sapling is kept at point P along AB so that $PA = \frac{1}{3} PB$. Find coordinates of P.

OPTIONS: (A) $P = (8, 0)$ (B) $P = (10, 0)$ (C) $P = (6, 0)$ (D) $P = (4, 0)$

ANSWER: (A) $P = (8, 0)$

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CHAPTER: 7 - TRIANGLES

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CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following is not the criterion for similarity of triangles?

OPTIONS: (A) AAA (B) SSS (C) SAS (D) RHS

ANSWER: (D) RHS

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the given figure, if $\triangle ABC \sim \triangle ADE$, then which of the following is true?

OPTIONS: (A) $AB/AC = AD/AE$ (B) $AB/AD = AC/AE$ (C) $AB/DE = AC/BC$ (D) $AB/AD = BC/DE$

ANSWER: (B) $AB/AD = AC/AE$

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Observe the given triangles. Which of the following statements is true?

OPTIONS: (A) $\triangle LMN \sim \triangle ABC$ (B) $\triangle LMN \sim \triangle ACB$ (C) $\triangle NML \sim \triangle ABC$ (D) $\triangle NML \sim \triangle ACB$

ANSWER: (C) $\triangle NML \sim \triangle ABC$

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Two isosceles triangles:

OPTIONS: (A) are always similar, but not congruent (B) are always congruent, but not similar
(C) may or may not be similar (D) are always similar and congruent

ANSWER: (C) may or may not be similar

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following statements is not always true?

OPTIONS: (A) Two circles are similar (B) Two isosceles right triangles are similar (C) Two rectangles are similar (D) Two equilateral triangles are similar

ANSWER: (C) Two rectangles are similar

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which types of triangles are always similar?

OPTIONS: (A) Right-angled triangles (B) Acute-angled triangles (C) Isosceles triangles (D) Equilateral triangles

ANSWER: (D) Equilateral triangles

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the given figure, if $DE \parallel BC$, $AD = 1.5$ cm, $DB = 3$ cm and $EC = 2$ cm, the length of AC is:

OPTIONS: (A) 1.5 cm (B) 3 cm (C) 3.5 cm (D) 4.5 cm

ANSWER: (C) 3.5 cm

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the given figure, if $DE \parallel BC$, $AD : AB = 1 : 3$ and $AE = 2.5$ cm, then AC equals:

OPTIONS: (A) 7.5 cm (B) 5 cm (C) 10 cm (D) 2.5 cm

ANSWER: (A) 7.5 cm

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the given figure, if $EA \parallel SR$ and $PE = x$ cm, then the value of $5x$ is:

OPTIONS: (A) 2.4 cm (B) 12 cm (C) 1.35 cm (D) 6.75 cm

ANSWER: (B) 12 cm

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the adjoining figure, if $\triangle AOB \sim \triangle COD$, then which of the following is true?

OPTIONS: (A) $AO.OB = OC.OD$ (B) $AO.CD = OC.AB$ (C) $AO.AB = OC.CD$ (D) $AO.OC = OB.OD$

ANSWER: (B) $AO.CD = OC.AB$

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the adjoining figure, if $\triangle PQT \sim \triangle RST$, which of the following is true?

OPTIONS: (A) $PQ/ST = QT/RS$ (B) $QT.RT = PT.ST$ (C) $PQ.PT = RS.RT$ (D) $PQ/PT = RT/RS$

ANSWER: (B) $QT.RT = PT.ST$

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: D is a point on side BC of $\triangle ABC$ such that $\angle ADC = \angle BAC$. Prove that $(CA)^2 = CB \times CD$.

OPTIONS: (A) $\triangle ADC \sim \triangle BAC$, so $CA/CD = CB/CA$ (B) $\triangle ADC \sim \triangle BAC$, so $CA/CD = CB/CA$ (C) $\triangle ADC \sim \triangle BAC$, so $CA/CD = CB/CA$ (D) $\triangle ADC \sim \triangle BAC$, so $CA/CD = CB/CA$

ANSWER: (A) $\triangle ADC \sim \triangle BAC$, so $CA/CD = CB/CA$

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CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: In the figure given below, $\angle 1 = \angle 2$ and $BE/BC = CD/AB$. Prove that $\triangle BDE \sim \triangle BAC$.

OPTIONS: (A) SAS similarity (B) AA similarity (C) SSS similarity (D) RHS similarity

ANSWER: (A) SAS similarity

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CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: In the given figure, if $PQ \parallel RS$, then prove that $\triangle POQ \sim \triangle SOR$.

OPTIONS: (A) AA similarity (B) SAS similarity (C) SSS similarity (D) RHS similarity

ANSWER: (A) AA similarity

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: In the given figure, $\triangle OSR \sim \triangle OQP$, $\angle ROQ = 125^\circ$ and $\angle ORS = 70^\circ$. Find the measures of $\angle OSR$ and $\angle OQP$.

OPTIONS: (A) $\angle OSR = 55^\circ$, $\angle OQP = 70^\circ$ (B) $\angle OSR = 55^\circ$, $\angle OQP = 55^\circ$ (C) $\angle OSR = 70^\circ$, $\angle OQP = 55^\circ$ (D) $\angle OSR = 70^\circ$, $\angle OQP = 70^\circ$

ANSWER: (A) $\angle OSR = 55^\circ$, $\angle OQP = 70^\circ$

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Two chords AB and CD intersect each other at point E. Prove that $\triangle AED \sim \triangle CEB$.

OPTIONS: (A) AA similarity (B) SAS similarity (C) SSS similarity (D) RHS similarity

ANSWER: (A) AA similarity

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CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: In the given figure, $\triangle PQR$ is a right triangle, right-angled at Q and $QS \perp PR$. Prove that $\triangle PQS \sim \triangle QRS$.

OPTIONS: (A) AA similarity (B) SAS similarity (C) SSS similarity (D) RHS similarity

ANSWER: (A) AA similarity

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Sides AB, AC and altitude AD of $\triangle ABC$ are proportional to sides PQ, PR and altitude PS of another triangle PQR. Prove that $\triangle ABC \sim \triangle PQR$.

OPTIONS: (A) SAS similarity (B) AA similarity (C) SSS similarity (D) RHS similarity

ANSWER: (A) SAS similarity

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CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: In the given figure, PQRS is a trapezium with $PQ \parallel SR$. Diagonals PR and QS intersect at T. Prove that $PT/TR = QT/TS$. Further, if $TS = TR$, prove that $\triangle PTS \sim \triangle QTR$.

OPTIONS: (A) AA similarity (B) SAS similarity (C) SSS similarity (D) RHS similarity

ANSWER: (A) AA similarity

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: (a) If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then prove that the other two sides are divided in the same ratio. (Basic Proportionality Theorem)

OPTIONS: (A) Statement and proof of BPT (B) Statement and proof of BPT (C) Statement and proof of BPT (D) Statement and proof of BPT

ANSWER: (A) Statement and proof of BPT

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CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: (b) AD and PS are respectively the medians of $\triangle ABC$ and $\triangle PQR$. If $\triangle ABC \sim \triangle PQR$, prove that (i) $\triangle ADC \sim \triangle PSR$ (ii) $AD/PS = BC/QR$.

OPTIONS: (A) SAS similarity (B) AA similarity (C) SSS similarity (D) RHS similarity

ANSWER: (A) SAS similarity

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: In the given figure, $\triangle ABC$ is right angled triangle with $\angle A = 90^\circ$. AD is perpendicular to BC. Prove that (i) $\triangle DBA \sim \triangle DAC$ (ii) $DA^2 = DB \times DC$ (iii) Find area when $DB = 9$ cm and $DC = 16$ cm.

OPTIONS: (A) $DA = 12$ cm, Area = 150 cm^2 (B) $DA = 12$ cm, Area = 150 cm^2 (C) $DA = 15$ cm, Area = 150 cm^2 (D) $DA = 12$ cm, Area = 120 cm^2

ANSWER: (A) $DA = 12$ cm, Area = 150 cm^2

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: State the SAS criteria of similarity of two triangles. In the given figure, it is given that $OA \times OC = OB \times OD$. Use SAS criteria to prove that $AD \parallel CB$.

OPTIONS: (A) $\triangle AOD \sim \triangle BOC$ (B) $\triangle AOD \sim \triangle BOC$ (C) $\triangle AOD \sim \triangle BOC$ (D) $\triangle AOD \sim \triangle BOC$

ANSWER: (A) $\triangle AOD \sim \triangle BOC$

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: State AA criterion of similarity of two triangles and use it to prove the following: In $\triangle ABC$ and $\triangle PQR$, AD and PS are angle bisectors of $\angle BAC$ and $\angle RPQ$ respectively. If $\triangle ABC \sim \triangle PQR$, prove that $\triangle ACD \sim \triangle PRS$.

OPTIONS: (A) AA similarity (B) SAS similarity (C) SSS similarity (D) RHS similarity

ANSWER: (A) AA similarity

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: State the converse of Basic Proportionality Theorem and use it to prove: Line segment joining mid-points of any two sides of a triangle is parallel to the third side.

OPTIONS: (A) Converse of BPT (B) Converse of BPT (C) Converse of BPT (D) Converse of BPT

ANSWER: (A) Converse of BPT

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Prove that in two concentric circles, the chord of the larger circle which touches the smaller circle is bisected at the point of contact. Hence, find the length of the chord if diameters are 10 cm and 8 cm.

OPTIONS: (A) 6 cm (B) 8 cm (C) 10 cm (D) 12 cm

ANSWER: (A) 6 cm

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: State "Basic Proportionality Theorem" and use it to prove: In a quadrilateral ABCD, diagonals AC and BD intersect at O such that $AO/BO = CO/DO$. Prove that ABCD is a trapezium.

OPTIONS: (A) $AB \parallel CD$ (B) $AD \parallel BC$ (C) $AB = CD$ (D) $AD = BC$

ANSWER: (A) $AB \parallel CD$

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: In the given figure, CM and RN are respectively the medians of $\triangle ABC$ and $\triangle PQR$. If $\triangle ABC \sim \triangle PQR$, prove that (i) $\triangle AMC \sim \triangle PNR$ (ii) $\angle BCM = \angle QRN$ (iii) $\triangle BMC \sim \triangle QNR$.

OPTIONS: (A) SAS similarity (B) AA similarity (C) SSS similarity (D) RHS similarity

ANSWER: (A) SAS similarity

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: PQRS is a trapezium with $PQ \parallel SR$. U and V are points on non-parallel sides PS and QR respectively. If $UV \parallel SR$, prove that $PU/US = QV/VR$.

OPTIONS: (A) Using BPT (B) Using BPT (C) Using BPT (D) Using BPT

ANSWER: (A) Using BPT

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: T is a point on the line PS produced of a parallelogram PQRS and QT intersects RS at V. Prove that $\triangle PQT \sim \triangle RVQ$.

OPTIONS: (A) AA similarity (B) SAS similarity (C) SSS similarity (D) RHS similarity

ANSWER: (A) AA similarity

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: In the given figure, $QR \parallel CB$ and $RP \parallel AC$. If $BR = 10$ cm, $QA = 12$ cm, $BP = 12$ cm and $PC = 18$ cm, find AR and QC .

OPTIONS: (A) $AR = 15$ cm, $QC = 8$ cm (B) $AR = 15$ cm, $QC = 8$ cm (C) $AR = 12$ cm, $QC = 10$ cm (D) $AR = 18$ cm, $QC = 6$ cm

ANSWER: (A) $AR = 15$ cm, $QC = 8$ cm

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The vertices of a $\triangle ABC$ are $A(-1, 3)$, $B(2, -3)$ and $C(4, 5)$. Find the coordinates of a point P on median AD such that $AP:PD = 2:3$.

OPTIONS: (A) $(3/5, 11/5)$ (B) $(3/5, 11/5)$ (C) $(3/5, 11/5)$ (D) $(3/5, 11/5)$

ANSWER: (A) $(3/5, 11/5)$

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: In $\triangle ABC$, D is a point on side BC such that $\angle ADC = \angle BAC$. Prove that $(CA)^2 = CB \times CD$.

OPTIONS: (A) $\triangle ADC \sim \triangle BAC$, so $CA/CD = CB/CA$ (B) $\triangle ADC \sim \triangle BAC$, so $CA/CD = CB/CA$ (C) $\triangle ADC \sim \triangle BAC$, so $CA/CD = CB/CA$ (D) $\triangle ADC \sim \triangle BAC$, so $CA/CD = CB/CA$

ANSWER: (A) $\triangle ADC \sim \triangle BAC$, so $CA/CD = CB/CA$

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: In the given figure, two triangles ABC and PQR are shown such that $\angle A = \angle P$ and $\angle C = \angle R$. If $AD \perp BC$ and $PS \perp QR$, prove that (i) $\triangle ADB \sim \triangle PSQ$ (ii) $AD \times QS = BD \times PS$.

OPTIONS: (A) AA similarity (B) SAS similarity (C) SSS similarity (D) RHS similarity

ANSWER: (A) AA similarity

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following graphs represents a polynomial with both zeroes being positive? (Diagram based, but the concept is about zeroes; I'll include the text version)

OPTIONS: (A) (i) only (B) (i) and (ii) (C) (ii) and (iii) (D) (ii) only

ANSWER: (D) (ii) only

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In which of the following graphs of the quadratic polynomials is the sum of zeroes 0?

OPTIONS: (A) (i) only (B) (i) and (ii) (C) (ii) and (iii) (D) (ii) only

ANSWER: (D) (ii) only

YEAR: 2026

CHAPTER: 7 - TRIANGLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In which of the following graphs of the polynomials is one of the zeroes twice the other?

OPTIONS: (A) (i) only (B) (i) and (ii) (C) (ii) and (iii) (D) (ii) only

ANSWER: (D) (ii) only

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CHAPTER: 8 - CIRCLES

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YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: PQ is tangent to a circle at a point P on the circle. The number of tangents which can be drawn to the circle parallel to PQ is:

OPTIONS: (A) 2 (B) 1 (C) many (D) zero

ANSWER: (B) 1

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In two concentric circles, a tangent to the smaller circle will intersect the larger circle at:

OPTIONS: (A) zero point (B) one point (C) two points (D) three points

ANSWER: (C) two points

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the given figure, PA is a tangent to a circle with centre O. If $OP = 10$ cm, then the length of AP is:

OPTIONS: (A) $10\sqrt{3}$ cm (B) 20 cm (C) 5 cm (D) $5\sqrt{3}$ cm

ANSWER: (D) $5\sqrt{3}$ cm

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the given figure, if AB is a tangent to the circle with centre O such that $OB = 6$ cm and $\angle AOB = 60^\circ$, then the length of OA is:

OPTIONS: (A) 3 cm (B) $3\sqrt{3}$ cm (C) $4\sqrt{3}$ cm (D) 12 cm

ANSWER: (B) $3\sqrt{3}$ cm

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the given figure, from which point can the tangent not be drawn to the circle with radius r ?

OPTIONS: (A) X (B) P (C) Q (D) Y

ANSWER: (A) X

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the given figure, PQ is tangent to the circle with centre O such that $OP = 2OQ$. The measure of $\angle OPQ$ is:

OPTIONS: (A) 15° (B) 60° (C) 45° (D) 30°

ANSWER: (D) 30°

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A chord QR subtends an angle of 105° at the centre O of the circle. The measure of $\angle RQP$ is:

OPTIONS: (A) $75^\circ/2$ (B) $105^\circ/2$ (C) 75° (D) 15°

ANSWER: (B) $105^\circ/2$

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: PT is tangent to the circle with centre O and radius 5 cm. OP intersects the circle at Q. If $PQ = x$, then PT^2 equals:

OPTIONS: (A) $x^2 + 5x$ (B) $x^2 + 10x + 50$ (C) $x^2 + 10x$ (D) $x^2 - 25$

ANSWER: (C) $x^2 + 10x$

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the given figure, PQ and PR are two tangents drawn to a circle with centre O. If $\angle ORQ = 25^\circ$, then the measure of $\angle PQR$ is:

OPTIONS: (A) 65° (B) 25° (C) 50° (D) 75°

ANSWER: (A) 65°

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Prove that the tangent drawn at any point of the circle is perpendicular to the radius through the point of contact.

OPTIONS: (A) Radius is perpendicular to tangent (B) Radius is perpendicular to tangent (C) Radius is perpendicular to tangent (D) Radius is perpendicular to tangent

ANSWER: (A) Radius is perpendicular to tangent

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Prove that the lengths of tangents drawn from an external point to a circle are equal.

OPTIONS: (A) Tangents from external point are equal (B) Tangents from external point are equal (C) Tangents from external point are equal (D) Tangents from external point are equal

ANSWER: (A) Tangents from external point are equal

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: In the given figure, AB is the diameter of the circle with centre O. Two tangents p and q are drawn at points A and B respectively. Prove that $p \parallel q$. Further, a line CD touches the circle at E and $\angle BCD = 110^\circ$. Find $\angle ADC$.

OPTIONS: (A) 70° (B) 80° (C) 90° (D) 60°

ANSWER: (A) 70°

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: A quadrilateral circumscribes the circle as shown. If $AB = 5$ cm, $BC = 7$ cm and $CD = 6$ cm, find AD .

OPTIONS: (A) 4 cm (B) 5 cm (C) 6 cm (D) 7 cm

ANSWER: (A) 4 cm

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Two tangents TP and TQ are drawn to a circle with centre O , from an external point T . Prove that $\angle PTQ = 2\angle OPQ$.

OPTIONS: (A) $\angle PTQ = 2\angle OPQ$ (B) $\angle PTQ = 2\angle OPQ$ (C) $\angle PTQ = 2\angle OPQ$ (D) $\angle PTQ = 2\angle OPQ$

ANSWER: (A) $\angle PTQ = 2\angle OPQ$

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that a parallelogram circumscribing a circle is a rhombus.

OPTIONS: (A) Parallelogram circumscribing circle is rhombus (B) Parallelogram circumscribing circle is rhombus (C) Parallelogram circumscribing circle is rhombus (D) Parallelogram circumscribing circle is rhombus

ANSWER: (A) Parallelogram circumscribing circle is rhombus

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that a rectangle circumscribing a circle is a square.

OPTIONS: (A) Rectangle circumscribing circle is square (B) Rectangle circumscribing circle is square (C) Rectangle circumscribing circle is square (D) Rectangle circumscribing circle is square

ANSWER: (A) Rectangle circumscribing circle is square

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Two tangents PA and PB are drawn from an external point P to a circle with centre O. If $\angle OAB = 15^\circ$, find the measures of each angle of $\triangle PAB$.

OPTIONS: (A) $\angle PAB = 75^\circ$, $\angle PBA = 75^\circ$, $\angle APB = 30^\circ$ (B) $\angle PAB = 75^\circ$, $\angle PBA = 75^\circ$, $\angle APB = 30^\circ$ (C) $\angle PAB = 75^\circ$, $\angle PBA = 75^\circ$, $\angle APB = 30^\circ$ (D) $\angle PAB = 75^\circ$, $\angle PBA = 75^\circ$, $\angle APB = 30^\circ$

ANSWER: (A) $\angle PAB = 75^\circ$, $\angle PBA = 75^\circ$, $\angle APB = 30^\circ$

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: In the given figure, TP and TQ are tangents at points P and Q of the circle respectively. If reflex $\angle POQ = 250^\circ$, find the measure of each angle of quadrilateral POQT.

OPTIONS: (A) $\angle POQ = 110^\circ$, $\angle PTQ = 70^\circ$ (B) $\angle POQ = 110^\circ$, $\angle PTQ = 70^\circ$ (C) $\angle POQ = 110^\circ$, $\angle PTQ = 70^\circ$ (D) $\angle POQ = 110^\circ$, $\angle PTQ = 70^\circ$

ANSWER: (A) $\angle POQ = 110^\circ$, $\angle PTQ = 70^\circ$

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Prove that in two concentric circles, the chord of the larger circle which touches the smaller circle is bisected at the point of contact. Hence, find the length of the chord if diameters are 10 cm and 8 cm.

OPTIONS: (A) 6 cm (B) 8 cm (C) 10 cm (D) 12 cm

ANSWER: (A) 6 cm

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: In the given figure, $\triangle ABC$ circumscribes a circle of radius 4 cm. If $AD = 7$ cm, $BD = 8$ cm, and area of $\triangle ABC = 84 \text{ cm}^2$, find BC and AC.

OPTIONS: (A) $BC = 14$ cm, $AC = 13$ cm (B) $BC = 14$ cm, $AC = 13$ cm (C) $BC = 15$ cm, $AC = 12$ cm (D) $BC = 13$ cm, $AC = 14$ cm

ANSWER: (A) $BC = 14$ cm, $AC = 13$ cm

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: In the given figure, quadrilateral PQRS circumscribes the circle with centre O. Prove that the opposite sides of the quadrilateral PQRS subtend supplementary angles at the centre O.

OPTIONS: (A) $\angle POQ + \angle ROS = 180^\circ$, $\angle QOR + \angle SOP = 180^\circ$ (B) $\angle POQ + \angle ROS = 180^\circ$, $\angle QOR + \angle SOP = 180^\circ$ (C) $\angle POQ + \angle ROS = 180^\circ$, $\angle QOR + \angle SOP = 180^\circ$ (D) $\angle POQ + \angle ROS = 180^\circ$, $\angle QOR + \angle SOP = 180^\circ$

ANSWER: (A) $\angle POQ + \angle ROS = 180^\circ$, $\angle QOR + \angle SOP = 180^\circ$

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Prove that opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle.

OPTIONS: (A) $\angle AOB + \angle COD = 180^\circ$, $\angle BOC + \angle DOA = 180^\circ$ (B) $\angle AOB + \angle COD = 180^\circ$, $\angle BOC + \angle DOA = 180^\circ$ (C) $\angle AOB + \angle COD = 180^\circ$, $\angle BOC + \angle DOA = 180^\circ$ (D) $\angle AOB + \angle COD = 180^\circ$, $\angle BOC + \angle DOA = 180^\circ$

ANSWER: (A) $\angle AOB + \angle COD = 180^\circ$, $\angle BOC + \angle DOA = 180^\circ$

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: In the given figure, a circle is inscribed in a quadrilateral ABCD which touches the sides AB, BC, CD and DA at P, Q, R and S respectively. Prove that $\angle AOB + \angle COD = 180^\circ$.

OPTIONS: (A) $\angle AOB + \angle COD = 180^\circ$ (B) $\angle AOB + \angle COD = 180^\circ$ (C) $\angle AOB + \angle COD = 180^\circ$ (D) $\angle AOB + \angle COD = 180^\circ$

ANSWER: (A) $\angle AOB + \angle COD = 180^\circ$

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: In the given figure, TP is tangent to a circle with centre O. Diameter BA when produced meets the tangent at T. If $\angle ABP = 35^\circ$, then find $\angle PTA$.

OPTIONS: (A) 20° (B) 30° (C) 40° (D) 50°

ANSWER: (A) 20°

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: In the given figure, AP, AQ and BC are tangents to the circle with centre O. If AB = 6 cm, AC = 7 cm and BC = 5 cm, find AP.

OPTIONS: (A) 9 cm (B) 10 cm (C) 8 cm (D) 7 cm

ANSWER: (A) 9 cm

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: In a circular museum hall of radius 14 m, statues are kept inside the inner concentric circle of radius 7 m. One statue in sector OAB is fenced along OA, AP, PB and BO where P is on outer circle. (i) Find $\angle AOP$. (ii) Prove $\triangle OAP \cong \triangle OBP$. (iii)(a) Find length of fencing required. (Take $\sqrt{3} = 1.73$)

OPTIONS: (A) $\angle AOP = 60^\circ$, fencing = 38.22 m (B) $\angle AOP = 60^\circ$, fencing = 38.22 m (C) $\angle AOP = 60^\circ$, fencing = 38.22 m (D) $\angle AOP = 60^\circ$, fencing = 38.22 m

ANSWER: (A) $\angle AOP = 60^\circ$, fencing = 38.22 m

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: In a circular museum hall of radius 14 m, statues are kept inside the inner concentric circle of radius 7 m. One statue in sector OAB is fenced along OA, AP, PB and BO where P is on outer circle. (i) Find $\angle AOP$. (ii) Prove $\triangle OAP \cong \triangle OBP$. (iii)(b) Find area of quadrilateral OAPB. (Take $\sqrt{3} = 1.73$)

OPTIONS: (A) Area = 84.77 m² (B) Area = 84.77 m² (C) Area = 84.77 m² (D) Area = 84.77 m²

ANSWER: (A) Area = 84.77 m²

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A farmer has a semicircular field with centre O and diameter 120 m. In segment ACPA he grows rice. In triangular region APB he grows vegetables. In segment BDPB cows graze. $\angle PAO = 30^\circ$ and $\angle POB = 60^\circ$. (i) Find radius and $\angle ABP$. (ii) Find area of sector AOP. (iii)(a) Find area of region where cows graze. (Use $\pi = 3.14$)

OPTIONS: (A) Radius = 60 m, $\angle ABP = 60^\circ$, Area = $1884 - 900\sqrt{3}$ m² (B) Radius = 60 m, $\angle ABP = 60^\circ$, Area = $1884 - 900\sqrt{3}$ m² (C) Radius = 60 m, $\angle ABP = 60^\circ$, Area = $1884 - 900\sqrt{3}$ m² (D) Radius = 60 m, $\angle ABP = 60^\circ$, Area = $1884 - 900\sqrt{3}$ m²

ANSWER: (A) Radius = 60 m, $\angle ABP = 60^\circ$, Area = $1884 - 900\sqrt{3} \text{ m}^2$

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A farmer has a semicircular field with centre O and diameter 120 m. In segment ACPA he grows rice. In triangular region APB he grows vegetables. In segment BDPB cows graze. $\angle PAO = 30^\circ$ and $\angle POB = 60^\circ$. (i) Find radius and $\angle ABP$. (ii) Find area of sector AOP. (iii)(b) Find area of region where vegetables are grown.

OPTIONS: (A) Area = $1800\sqrt{3} \text{ m}^2$ (B) Area = $1800\sqrt{3} \text{ m}^2$ (C) Area = $1800\sqrt{3} \text{ m}^2$ (D) Area = $1800\sqrt{3} \text{ m}^2$

ANSWER: (A) Area = $1800\sqrt{3} \text{ m}^2$

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Two concentric circles are of radii 6 cm and 10 cm. Find the length of the chord of the larger circle which touches the smaller circle.

OPTIONS: (A) 16 cm (B) 14 cm (C) 12 cm (D) 18 cm

ANSWER: (A) 16 cm

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: In the given figure, $AB \parallel DC$. If $OB = 3OD$ and $CD = 1.8 \text{ cm}$, find AB.

OPTIONS: (A) 5.4 cm (B) 5.4 cm (C) 6.0 cm (D) 4.8 cm

ANSWER: (A) 5.4 cm

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: In the given figure, O is the centre of a circle of radius 7 cm. AB is a chord of the circle. Find the perimeter of the shaded region.

OPTIONS: (A) 44 cm (B) 44 cm (C) 44 cm (D) 44 cm

ANSWER: (A) 44 cm

YEAR: 2026

CHAPTER: 8 - CIRCLES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: In the given figure, PQ is a tangent to a circle with centre O(-5, 3). If coordinates of P and Q are (3, 1) and (0, 6) respectively, then using distance formula, show that $PQ \perp OQ$.

OPTIONS: (A) $PQ \perp OQ$ (B) $PQ \perp OQ$ (C) $PQ \perp OQ$ (D) $PQ \perp OQ$

ANSWER: (A) $PQ \perp OQ$

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CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

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YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following statements is false? (A) $\tan 45^\circ = \cot 45^\circ$ (B) $\sin 90^\circ = \tan 45^\circ$ (C) $\sin 30^\circ = \cos 30^\circ$ (D) $\sin 45^\circ = \cos 45^\circ$

OPTIONS: (A) $\tan 45^\circ = \cot 45^\circ$ (B) $\sin 90^\circ = \tan 45^\circ$ (C) $\sin 30^\circ = \cos 30^\circ$ (D) $\sin 45^\circ = \cos 45^\circ$

ANSWER: (C) $\sin 30^\circ = \cos 30^\circ$

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of $(\tan^2 A - 1/\cos^2 A)$ is:

OPTIONS: (A) more than 1 (B) 1 (C) 0 (D) -1

ANSWER: (D) -1

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of $(\cot^2 A - 1/\sin^2 A)$ is:

OPTIONS: (A) more than 1 (B) 1 (C) 0 (D) -1

ANSWER: (D) -1

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of $(1/\sec^2 A + 1/\operatorname{cosec}^2 A)$ is:

OPTIONS: (A) more than 1 (B) 1 (C) 0 (D) -1

ANSWER: (B) 1

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of $(\sec^2 30^\circ + \tan^2 30^\circ)/(\sin^2 45^\circ + \cos^2 45^\circ)$ is:

OPTIONS: (A) 1 (B) $5/3$ (C) $13/3$ (D) 7

ANSWER: (C) $13/3$

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of $2 \tan 45^\circ \sin^2 60^\circ - \cos 90^\circ$ is:

OPTIONS: (A) $-1/4$ (B) $3/2$ (C) $3/4$ (D) $1/4$

ANSWER: (B) $3/2$

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of $\sin 90^\circ \cos 90^\circ - \sin^2 60^\circ$ is:

OPTIONS: (A) $1/4$ (B) $3/4$ (C) $-3/4$ (D) $5/4$

ANSWER: (C) $-3/4$

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of $3 \cos 30^\circ - 4 \cos^3 30^\circ$ is same as the value of:

OPTIONS: (A) $\sin 90^\circ$ (B) $\cos 90^\circ$ (C) $\tan 90^\circ$ (D) $\sec 90^\circ$

ANSWER: (B) $\cos 90^\circ$

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of $\tan^2 60^\circ - 1 / \tan^2 60^\circ + 1$ is equal to the value of:

OPTIONS: (A) $\tan 60^\circ$ (B) $\cos 60^\circ$ (C) $\sec 60^\circ$ (D) $\sin 60^\circ$

ANSWER: (B) $\cos 60^\circ$

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of $\tan 60^\circ - \cot 60^\circ$ is equal to the value of:

OPTIONS: (A) $\sec 30^\circ$ (B) $\operatorname{cosec} 30^\circ$ (C) $\cos 30^\circ$ (D) $\sin 60^\circ$

ANSWER: (A) $\sec 30^\circ$

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of $2 \sin 45^\circ \cos 45^\circ$ is same as the value of:

OPTIONS: (A) $\sin 0^\circ$ (B) $\cos 0^\circ$ (C) $\tan 90^\circ$ (D) $\cot 90^\circ$

ANSWER: (B) $\cos 0^\circ$

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $\sqrt{3} \sin A = \cos A$, then the measure of A is:

OPTIONS: (A) 90° (B) 60° (C) 45° (D) 30°

ANSWER: (D) 30°

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $\tan \theta - \cot \theta = 0$, then the value of θ is:

OPTIONS: (A) 30° (B) 45° (C) 60° (D) 90°

ANSWER: (B) 45°

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $2 \sin 2\theta = \sqrt{2}$, then the value of θ is:

OPTIONS: (A) 90° (B) 60° (C) 45° (D) 22.5°

ANSWER: (D) 22.5°

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $\sin \theta = 1/7$, then $\tan \theta$ is:

OPTIONS: (A) $1/(4\sqrt{3})$ (B) $1/(2\sqrt{3})$ (C) $4\sqrt{3}/7$ (D) $6/7$

ANSWER: (A) $1/(4\sqrt{3})$

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $\cot \theta = \sqrt{5}$, then $\sin \theta$ equals:

OPTIONS: (A) $1/\sqrt{6}$ (B) $\sqrt{6}$ (C) $\sqrt{5}/6$ (D) $1/2$

ANSWER: (A) $1/\sqrt{6}$

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $\sin \theta = 1/\sqrt{11}$, then $\cot \theta$ equals:

OPTIONS: (A) $\sqrt{11}/\sqrt{10}$ (B) $\sqrt{10}/\sqrt{11}$ (C) $\sqrt{10}$ (D) $\sqrt{11}$

ANSWER: (C) $\sqrt{10}$

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If $\sec \theta = \sqrt{3}$, then $\tan \theta$ equals:

OPTIONS: (A) $\sqrt{3} - 1$ (B) 60° (C) 2 (D) $\sqrt{2}$

ANSWER: (D) $\sqrt{2}$

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of $(\sec 45^\circ - \cos 45^\circ)$ is equal to:

OPTIONS: (A) $\sin 45^\circ$ (B) $\sin 90^\circ$ (C) $\cos 90^\circ$ (D) $\tan 45^\circ$

ANSWER: (A) $\sin 45^\circ$

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following is not defined for $x = 90^\circ$?

OPTIONS: (A) $\cot x$ (B) $\operatorname{cosec} x$ (C) $\tan(x/2)$ (D) $\sec x$

ANSWER: (D) $\sec x$

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Evaluate: $\cos 45^\circ / (\sec 30^\circ + \operatorname{cosec} 30^\circ)$

OPTIONS: (A) $(3 - \sqrt{3})/(4\sqrt{2})$ (B) $(3 - \sqrt{3})/(4\sqrt{2})$ (C) $(3 + \sqrt{3})/(4\sqrt{2})$ (D) $(3 - \sqrt{3})/(2\sqrt{2})$

ANSWER: (A) $(3 - \sqrt{3})/(4\sqrt{2})$

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Evaluate: $\sin 45^\circ / (\sec 30^\circ - \tan 30^\circ)$

OPTIONS: (A) $(3 + \sqrt{3})/(4\sqrt{2})$ (B) $(3 + \sqrt{3})/(4\sqrt{2})$ (C) $(3 - \sqrt{3})/(4\sqrt{2})$ (D) $(3 + \sqrt{3})/(2\sqrt{2})$

ANSWER: (A) $(3 + \sqrt{3})/(4\sqrt{2})$

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Evaluate: $(5 \sin^2 45^\circ - 3 \tan^2 30^\circ) / (2 \sec^2 30^\circ)$

OPTIONS: (A) $9/16$ (B) $9/16$ (C) $9/16$ (D) $9/16$

ANSWER: (A) $9/16$

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Evaluate: $(4 \cos^3 60^\circ + \operatorname{cosec} 30^\circ) / \tan^2 30^\circ$

OPTIONS: (A) $15/2$ (B) $15/2$ (C) $15/2$ (D) $15/2$

ANSWER: (A) $15/2$

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If $\sin 3A = 1$, find $\cos 2A - \tan^2 45^\circ$.

OPTIONS: (A) $-1/2$ (B) $-1/2$ (C) $-1/2$ (D) $-1/2$

ANSWER: (A) $-1/2$

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If $(\sec A + \tan A)(1 - \sin A) = k \cos A$, find k .

OPTIONS: (A) 1 (B) 1 (C) 1 (D) 1

ANSWER: (A) 1

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If $\sin A = 1/2$ and $\tan B = \sqrt{3}$, verify that $\cos(A+B) = \cos A \cos B - \sin A \sin B$.

OPTIONS: (A) LHS = RHS (B) LHS = RHS (C) LHS = RHS (D) LHS = RHS

ANSWER: (A) LHS = RHS

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: For $A = 60^\circ$ and $B = 30^\circ$, verify that $\tan(A-B) = (\tan A - \tan B)/(1 + \tan A \tan B)$.

OPTIONS: (A) LHS = RHS (B) LHS = RHS (C) LHS = RHS (D) LHS = RHS

ANSWER: (A) LHS = RHS

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $(\sin A - \operatorname{cosec} A)^2 + (\cos A - \sec A)^2 = \tan^2 A + \cot^2 A - 1$

OPTIONS: (A) LHS = RHS (B) LHS = RHS (C) LHS = RHS (D) LHS = RHS

ANSWER: (A) LHS = RHS

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $(1 + \cot^2 A)/(1 + \tan^2 A) = ((1 - \cot A)/(1 - \tan A))^2$

OPTIONS: (A) LHS = RHS (B) LHS = RHS (C) LHS = RHS (D) LHS = RHS

ANSWER: (A) LHS = RHS

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $\sqrt{(\operatorname{cosec} A - 1)/(\operatorname{cosec} A + 1)} = \sec A - \tan A$

OPTIONS: (A) LHS = RHS (B) LHS = RHS (C) LHS = RHS (D) LHS = RHS

ANSWER: (A) LHS = RHS

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CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $\sqrt{(1 - \sin A)/(1 + \sin A)} = 1/(\sec A + \tan A)$

OPTIONS: (A) LHS = RHS (B) LHS = RHS (C) LHS = RHS (D) LHS = RHS

ANSWER: (A) LHS = RHS

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CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $(1 + \cos A)/\cos A = \cos^2 A/(1 - \sin A)$

OPTIONS: (A) LHS = RHS (B) LHS = RHS (C) LHS = RHS (D) LHS = RHS

ANSWER: (A) LHS = RHS

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CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $(1 + \sec A)/\sec A = \sin^2 A/(1 - \cos A)$

OPTIONS: (A) LHS = RHS (B) LHS = RHS (C) LHS = RHS (D) LHS = RHS

ANSWER: (A) LHS = RHS

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $\cot A - \cos A / \cot A + \cos A = \sec A - \tan A / \sec A + \tan A$

OPTIONS: (A) LHS = RHS (B) LHS = RHS (C) LHS = RHS (D) LHS = RHS

ANSWER: (A) LHS = RHS

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $(\operatorname{cosec} A - \sin A)(\sec A - \cos A) = \tan A + \cot A$

OPTIONS: (A) LHS = RHS (B) LHS = RHS (C) LHS = RHS (D) LHS = RHS

ANSWER: (A) LHS = RHS

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $\cos \theta / (1 + \sin \theta) + (1 + \sin \theta) / \cos \theta = 2 \sec \theta$

OPTIONS: (A) LHS = RHS (B) LHS = RHS (C) LHS = RHS (D) LHS = RHS

ANSWER: (A) LHS = RHS

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $(1 - \sin \theta) / \cos \theta + \cos \theta / (1 - \sin \theta) = 2 \sec \theta$

OPTIONS: (A) LHS = RHS (B) LHS = RHS (C) LHS = RHS (D) LHS = RHS

ANSWER: (A) LHS = RHS

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $\tan^2 \theta + \cot^2 \theta + 2 = \sec^2 \theta \operatorname{cosec}^2 \theta$

OPTIONS: (A) LHS = RHS (B) LHS = RHS (C) LHS = RHS (D) LHS = RHS
ANSWER: (A) LHS = RHS

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $\tan \theta / (1 + \cot \theta) + \cot \theta / (1 + \tan \theta) = \tan \theta + \cot \theta - 1$

OPTIONS: (A) LHS = RHS (B) LHS = RHS (C) LHS = RHS (D) LHS = RHS

ANSWER: (A) LHS = RHS

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $\cos A - 2 \cos^3 A / 2 \sin^3 A - \sin A = \cot A$

OPTIONS: (A) LHS = RHS (B) LHS = RHS (C) LHS = RHS (D) LHS = RHS

ANSWER: (A) LHS = RHS

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $\sin A - \tan A / \sin A + \tan A = 1 - \sec A / 1 + \sec A$

OPTIONS: (A) LHS = RHS (B) LHS = RHS (C) LHS = RHS (D) LHS = RHS

ANSWER: (A) LHS = RHS

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $(\sec \theta + \tan \theta)^2 - 1 / (\sec \theta + \tan \theta)^2 + 1 = \sin \theta$

OPTIONS: (A) LHS = RHS (B) LHS = RHS (C) LHS = RHS (D) LHS = RHS

ANSWER: (A) LHS = RHS

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $\tan A/(1-\cot A) + \cot A/(1-\tan A) = 1 + \sec A \operatorname{cosec} A$

OPTIONS: (A) LHS = RHS (B) LHS = RHS (C) LHS = RHS (D) LHS = RHS

ANSWER: (A) LHS = RHS

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $\sqrt{(1 - \cos A)/(1 + \cos A)} = \tan A/(\sec A + 1)$

OPTIONS: (A) LHS = RHS (B) LHS = RHS (C) LHS = RHS (D) LHS = RHS

ANSWER: (A) LHS = RHS

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CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: If $\sin(A + 2B) = 1$ and $\cos(2A + B) = 1/2$, find A and B. Hence find $\tan(B - A)$.

OPTIONS: (A) $A = 10^\circ$, $B = 40^\circ$, $\tan(30^\circ) = 1/\sqrt{3}$ (B) $A = 10^\circ$, $B = 40^\circ$, $\tan(30^\circ) = 1/\sqrt{3}$ (C) $A = 10^\circ$, $B = 40^\circ$, $\tan(30^\circ) = 1/\sqrt{3}$ (D) $A = 10^\circ$, $B = 40^\circ$, $\tan(30^\circ) = 1/\sqrt{3}$

ANSWER: (A) $A = 10^\circ$, $B = 40^\circ$, $\tan(30^\circ) = 1/\sqrt{3}$

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CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: If $\sin x = p$, prove that (i) $\cot x = \sqrt{(1 - p^2)}/p$ (ii) $(1 + \tan^2 x)/(1 + \cot^2 x) = p^2/(1 - p^2)$

OPTIONS: (A) (i) and (ii) are proven (B) (i) and (ii) are proven (C) (i) and (ii) are proven (D) (i) and (ii) are proven

ANSWER: (A) (i) and (ii) are proven

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Triangle ABC is an isosceles right triangle, right angled at B. Find $\sin^2 A + \cos^2 C$.

OPTIONS: (A) 1 (B) 1 (C) 1 (D) 1

ANSWER: (A) 1

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Triangle PQR is an isosceles right triangle, right angled at Q. Find $\sec^2 P + \operatorname{cosec}^2 R$.

OPTIONS: (A) 4 (B) 4 (C) 4 (D) 4

ANSWER: (A) 4

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Triangle XYZ is an isosceles right triangle, right angled at X. Find $\operatorname{cosec}^2 Y + \tan^2 Z$.

OPTIONS: (A) 3 (B) 3 (C) 3 (D) 3

ANSWER: (A) 3

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Evaluate: $\sin 45^\circ \cos 45^\circ + \operatorname{cosec}^2 30^\circ$

OPTIONS: (A) $9/2$ (B) $9/2$ (C) $9/2$ (D) $9/2$

ANSWER: (A) $9/2$

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Evaluate: $\sec 60^\circ \cos^2 30^\circ + \sin^2 45^\circ$

OPTIONS: (A) 2 (B) 2 (C) 2 (D) 2

ANSWER: (A) 2

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CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Evaluate: $(2 \sin^2 60^\circ + \cos^2 60^\circ) / \tan^2 30^\circ$

OPTIONS: (A) $7/2$ (B) $7/2$ (C) $7/2$ (D) $7/2$

ANSWER: (A) $7/2$

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CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Evaluate: $\sin^2 30^\circ - \cos^2 45^\circ + \cot^2 60^\circ$
OPTIONS: (A) $1/12$ (B) $1/12$ (C) $1/12$ (D) $1/12$
ANSWER: (A) $1/12$

YEAR: 2026
CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Evaluate: $\sin^2 45^\circ / (\operatorname{cosec}^2 30^\circ - \tan^2 45^\circ)$
OPTIONS: (A) $1/6$ (B) $1/6$ (C) $1/6$ (D) $1/6$
ANSWER: (A) $1/6$

YEAR: 2026
CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Evaluate: $(4 \cos^3 60^\circ + \operatorname{cosec} 30^\circ) / \tan^2 30^\circ$
OPTIONS: (A) $15/2$ (B) $15/2$ (C) $15/2$ (D) $15/2$
ANSWER: (A) $15/2$

YEAR: 2026
CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: If $\sin 2A = \sqrt{3}/2$ and $2 \tan B + 1 = 3$, find $(A+B)$.
OPTIONS: (A) 75° (B) 75° (C) 75° (D) 75°
ANSWER: (A) 75°

YEAR: 2026
CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: If $\sin(A-B) = 1/2$ and $\tan(A+B) = \sqrt{3}$ ($0^\circ \leq A+B < 90^\circ$ and $A > B$), find A and B.

OPTIONS: (A) $A = 37.5^\circ$, $B = 7.5^\circ$ (B) $A = 37.5^\circ$, $B = 7.5^\circ$ (C) $A = 37.5^\circ$, $B = 7.5^\circ$ (D) $A = 37.5^\circ$, $B = 7.5^\circ$

ANSWER: (A) $A = 37.5^\circ$, $B = 7.5^\circ$

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: If $\sin(A+2B) = 1$ and $A = 3B$, find A and B.

OPTIONS: (A) $A = 54^\circ$, $B = 18^\circ$ (B) $A = 54^\circ$, $B = 18^\circ$ (C) $A = 54^\circ$, $B = 18^\circ$ (D) $A = 54^\circ$, $B = 18^\circ$

ANSWER: (A) $A = 54^\circ$, $B = 18^\circ$

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: If $\sin A + \sin^2 A = 1$, find $\cos^2 A + \cos^4 A$. Also prove $\tan^2 A \sec^2 A = 1$.

OPTIONS: (A) $\cos^2 A + \cos^4 A = 1$; $\tan^2 A \sec^2 A = 1$ (B) $\cos^2 A + \cos^4 A = 1$; $\tan^2 A \sec^2 A = 1$
(C) $\cos^2 A + \cos^4 A = 1$; $\tan^2 A \sec^2 A = 1$ (D) $\cos^2 A + \cos^4 A = 1$; $\tan^2 A \sec^2 A = 1$

ANSWER: (A) $\cos^2 A + \cos^4 A = 1$; $\tan^2 A \sec^2 A = 1$

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $(1 + \cos \theta)/\cos \theta = \cos^2 \theta/(1 - \sin \theta)$

OPTIONS: (A) LHS = RHS (B) LHS = RHS (C) LHS = RHS (D) LHS = RHS

ANSWER: (A) LHS = RHS

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Prove that: $(\operatorname{cosec} A - \sin A)(\sec A - \cos A) = 1/(\tan A + \cot A)$

OPTIONS: (A) LHS = RHS (B) LHS = RHS (C) LHS = RHS (D) LHS = RHS

ANSWER: (A) LHS = RHS

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CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: One of the possible values of A, for which $\cos 2A = \cos A$, is:

OPTIONS: (A) 0° (B) 30° (C) 45° (D) 90°

ANSWER: (A) 0°

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The value of $(\cos 90^\circ - \sin 90^\circ)$ is:

OPTIONS: (A) -1 (B) greater than 0 (C) equal to $\tan 45^\circ$ (D) 0

ANSWER: (A) -1

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): For an acute angle θ , $\cos \theta$ is always less than 1. Reason (R): In a right-angled triangle, hypotenuse is the longest side and $\cos \theta = \text{Base}/\text{Hypotenuse}$.

OPTIONS: (A) Both A and R true, R explains A (B) Both A and R true, R does not explain A (C) A true, R false (D) A false, R true

ANSWER: (A) Both A and R true, R explains A

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): For $\sin \theta = 1$, $\cos \theta$ must be 0. Reason (R): $\sin^2 \theta - \cos^2 \theta = 1$.

OPTIONS: (A) Both A and R true, R explains A (B) Both A and R true, R does not explain A (C) A true, R false (D) A false, R true

ANSWER: (C) A true, R false

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): For an acute angle θ , $\cot \theta = 1 \Rightarrow \operatorname{cosec} \theta = 2$. Reason (R): $\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$.

OPTIONS: (A) Both A and R true, R explains A (B) Both A and R true, R does not explain A (C) A true, R false (D) A false, R true
ANSWER: (A) Both A and R true, R explains A

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): $(2 - \sqrt{5})^2$ is a rational number. Reason (R): $(\sqrt{5})^2$ is a rational number.

OPTIONS: (A) Both A and R true, R explains A (B) Both A and R true, R does not explain A (C) A true, R false (D) A false, R true

ANSWER: (D) A false, R true

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): $(3 + 2\sqrt{2})^2$ is a rational number. Reason (R): $\sqrt{2}$ is an irrational number.

OPTIONS: (A) Both A and R true, R explains A (B) Both A and R true, R does not explain A (C) A true, R false (D) A false, R true

ANSWER: (D) A false, R true

YEAR: 2026

CHAPTER: 9 - INTRODUCTION TO TRIGONOMETRY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): $(\sqrt{3} + 1)^2$ is a rational number. Reason (R): $(\sqrt{3})^2 = 3$ is a rational number.

OPTIONS: (A) Both A and R true, R explains A (B) Both A and R true, R does not explain A (C) A true, R false (D) A false, R true

ANSWER: (D) A false, R true

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CHAPTER: 10 - HEIGHTS AND DISTANCES

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YEAR: 2026

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: At an instant, the length of shadow of a stick is found to be $\sqrt{3}$ times the length of the stick. The Sun's altitude at that instant is:

OPTIONS: (A) 30° (B) 45° (C) 60° (D) 90°

ANSWER: (A) 30°

YEAR: 2026

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The length of the shadow of a tower when the sun's altitude changes from 30° to 60° will:

OPTIONS: (A) become shorter (B) become longer (C) remain same (D) be doubled

ANSWER: (A) become shorter

YEAR: 2026

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the given figure, the angle of depression of the foot of building CD from the top of tower AB is:

OPTIONS: (A) 60° (B) 30° (C) 45° (D) 90°

ANSWER: (A) 60°

YEAR: 2026

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the given figure, the angle of elevation of point A from point C is:

OPTIONS: (A) 30° (B) 45° (C) 60° (D) Cannot be determined

ANSWER: (C) 60°

YEAR: 2026

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: A vertical pole of height 10 m casts a shadow of 15 m on the ground and at the same time, a tower casts a shadow of 45 m on the ground. Find the height of the tower.

OPTIONS: (A) 30 m (B) 30 m (C) 30 m (D) 30 m

ANSWER: (A) 30 m

YEAR: 2026

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: An injured bird was found on the roof of a 15 m high building. A fireman used a ladder making an angle of 60° with the ground. (i) Find length of ladder. (ii) Find distance from building. (iii)(a) If ladder is placed opposite wall making 30° with ground, draw diagram and find width of road.

OPTIONS: (A) Ladder = $10\sqrt{3}$ m, Distance = $5\sqrt{3}$ m, Road width = $15\sqrt{3}$ m (B) Ladder = $10\sqrt{3}$ m, Distance = $5\sqrt{3}$ m, Road width = $15\sqrt{3}$ m (C) Ladder = $10\sqrt{3}$ m, Distance = $5\sqrt{3}$ m, Road width = $15\sqrt{3}$ m (D) Ladder = $10\sqrt{3}$ m, Distance = $5\sqrt{3}$ m, Road width = $15\sqrt{3}$ m

ANSWER: (A) Ladder = $10\sqrt{3}$ m, Distance = $5\sqrt{3}$ m, Road width = $15\sqrt{3}$ m

YEAR: 2026

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: An injured bird was found on the roof of a 15 m high building. A fireman used a ladder making an angle of 60° with the ground. (i) Find length of ladder. (ii) Find distance from building. (iii)(b) If ladder is placed opposite wall making 30° with ground, find length of ladder.

OPTIONS: (A) Ladder = $10\sqrt{3}$ m, Distance = $5\sqrt{3}$ m, New Ladder = 30 m (B) Ladder = $10\sqrt{3}$ m, Distance = $5\sqrt{3}$ m, New Ladder = 30 m (C) Ladder = $10\sqrt{3}$ m, Distance = $5\sqrt{3}$ m, New Ladder = 30 m (D) Ladder = $10\sqrt{3}$ m, Distance = $5\sqrt{3}$ m, New Ladder = 30 m

ANSWER: (A) Ladder = $10\sqrt{3}$ m, Distance = $5\sqrt{3}$ m, New Ladder = 30 m

YEAR: 2026

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A flagstaff 7.32 m long is fitted at the top of a 10 m tall building. Ropes are tied from point P on the ground to base and top of flagstaff. Rope I1 makes 30° with horizontal and I2 makes angle θ with horizontal. (i) Find x (distance from base). (ii) Find θ . (iii)(a) Find total length of ropes needed. (Use $\sqrt{2} = 1.4$, $\sqrt{3} = 1.732$)

OPTIONS: (A) $x = 17.32$ m, $\theta = 45^\circ$, Total = 44.248 m (B) $x = 17.32$ m, $\theta = 45^\circ$, Total = 44.248 m (C) $x = 17.32$ m, $\theta = 45^\circ$, Total = 44.248 m (D) $x = 17.32$ m, $\theta = 45^\circ$, Total = 44.248 m

ANSWER: (A) $x = 17.32$ m, $\theta = 45^\circ$, Total = 44.248 m

YEAR: 2026

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A flagstaff 7.32 m long is fitted at the top of a 10 m tall building. Ropes are tied from point P on the ground to base and top of flagstaff. Rope I1 makes 30° with horizontal and I2 makes angle θ with horizontal. (i) Find x (distance from base). (ii) Find θ . (iii)(b) Which rope is longer I1 or I2 and by how much? (Use $\sqrt{2} = 1.4$, $\sqrt{3} = 1.732$)

OPTIONS: (A) I2 is longer by 4.248 m (B) I2 is longer by 4.248 m (C) I2 is longer by 4.248 m (D) I2 is longer by 4.248 m

ANSWER: (A) I2 is longer by 4.248 m

YEAR: 2026

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: Reena's kite is at height 60 m above ground. String is tied to a point on ground making 30° inclination. Ravi is flying from a 10 m high building, his kite is also at

60 m height and string length same as Reena's. θ is angle of elevation from rooftop. (i) Find length of string used by Reena. (ii) Find $\sin \theta$. (iii)(a) If θ changes to 60° without changing string length, find height of Ravi's kite above ground. (Use $\sqrt{3} = 1.7$)

OPTIONS: (A) String = 120 m, $\sin \theta = 5/12$, New height = 61 m (B) String = 120 m, $\sin \theta = 5/12$, New height = 61 m (C) String = 120 m, $\sin \theta = 5/12$, New height = 61 m (D) String = 120 m, $\sin \theta = 5/12$, New height = 61 m

ANSWER: (A) String = 120 m, $\sin \theta = 5/12$, New height = 61 m

YEAR: 2026

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: Reena's kite is at height 60 m above ground. String is tied to a point on ground making 30° inclination. Ravi is flying from a 10 m high building, his kite is also at 60 m height and string length same as Reena's. θ is angle of elevation from rooftop. (i) Find length of string used by Reena. (ii) Find $\sin \theta$. (iii)(b) If string had inclination of 30° with ground, what would be height of Ravi's kite?

OPTIONS: (A) String = 120 m, $\sin \theta = 5/12$, Height = 70 m (B) String = 120 m, $\sin \theta = 5/12$, Height = 70 m (C) String = 120 m, $\sin \theta = 5/12$, Height = 70 m (D) String = 120 m, $\sin \theta = 5/12$, Height = 70 m

ANSWER: (A) String = 120 m, $\sin \theta = 5/12$, Height = 70 m

YEAR: 2026

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A tree broke during a storm such that its broken part formed an angle of 30° with the ground. The distance from base to where top touches the ground is 10 m. (i) Draw neat diagram. (ii) Find height above ground at which tree broke. (iii)(a) Find height of tree before it broke. (Use $\sqrt{3} = 1.732$)

OPTIONS: (A) Height of break = $10/\sqrt{3}$ m, Total height = $10\sqrt{3}$ m = 17.32 m (B) Height of break = $10/\sqrt{3}$ m, Total height = $10\sqrt{3}$ m = 17.32 m (C) Height of break = $10/\sqrt{3}$ m, Total height = $10\sqrt{3}$ m = 17.32 m (D) Height of break = $10/\sqrt{3}$ m, Total height = $10\sqrt{3}$ m = 17.32 m

ANSWER: (A) Height of break = $10/\sqrt{3}$ m, Total height = $10\sqrt{3}$ m = 17.32 m

YEAR: 2026

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A tree broke during a storm such that its broken part formed an angle of 30° with the ground. The distance from base to where top touches the ground is 10 m. (i) Draw neat diagram. (ii) Find height above ground at which tree broke. (iii)(b) If another tree broke from the same height but broken part made 60° angle with ground, find total height.

OPTIONS: (A) Height of break = $10/\sqrt{3}$ m, Total height = $10(2+\sqrt{3})/3$ m = 12.44 m (B) Height of break = $10/\sqrt{3}$ m, Total height = $10(2+\sqrt{3})/3$ m = 12.44 m (C) Height of break = $10/\sqrt{3}$ m, Total height = $10(2+\sqrt{3})/3$ m = 12.44 m (D) Height of break = $10/\sqrt{3}$ m, Total height = $10(2+\sqrt{3})/3$ m = 12.44 m

ANSWER: (A) Height of break = $10/\sqrt{3}$ m, Total height = $10(2+\sqrt{3})/3$ m = 12.44 m

YEAR: 2026

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The angle of elevation of the top of a building from the foot of a tower is 30° and the angle of elevation of the top of the tower from the foot of the building is 60° . If the tower is 60 m high, find the height of the building and distance between them. (Use $\sqrt{3} = 1.73$)

OPTIONS: (A) Building height = 20 m, Distance = 34.6 m (B) Building height = 20 m, Distance = 34.6 m (C) Building height = 20 m, Distance = 34.6 m (D) Building height = 20 m, Distance = 34.6 m

ANSWER: (A) Building height = 20 m, Distance = 34.6 m

YEAR: 2026

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A straight road leads to foot of a tower. The shadow is found to be 40 m longer when sun's altitude is 30° than when it is 60° . Find the height of the tower and length of shadow in both situations. (Use $\sqrt{3} = 1.73$)

OPTIONS: (A) Height = 34.6 m, Shadow at 60° = 20 m, Shadow at 30° = 60 m (B) Height = 34.6 m, Shadow at 60° = 20 m, Shadow at 30° = 60 m (C) Height = 34.6 m, Shadow at

$60^\circ = 20$ m, Shadow at $30^\circ = 60$ m (D) Height = 34.6 m, Shadow at $60^\circ = 20$ m, Shadow at $30^\circ = 60$ m

ANSWER: (A) Height = 34.6 m, Shadow at $60^\circ = 20$ m, Shadow at $30^\circ = 60$ m

YEAR: 2026

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Two poles of equal heights are standing opposite each other on either side of a road which is 80 m wide. From a point between them, the angles of elevation of the top of the poles are 60° and 30° respectively. Find the height of the poles and the distance of the point from the poles.

OPTIONS: (A) Height = $20\sqrt{3}$ m, Distances = 20 m and 60 m (B) Height = $20\sqrt{3}$ m, Distances = 20 m and 60 m (C) Height = $20\sqrt{3}$ m, Distances = 20 m and 60 m (D) Height = $20\sqrt{3}$ m, Distances = 20 m and 60 m

ANSWER: (A) Height = $20\sqrt{3}$ m, Distances = 20 m and 60 m

YEAR: 2026

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A statue 2 m tall stands on the top of a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and from the same point the angle of elevation of the bottom of the statue is 45° . Find the height of the pedestal and its distance from the point. (Use $\sqrt{3} = 1.73$)

OPTIONS: (A) Height of pedestal = 2.73 m, Distance = 2.73 m (B) Height of pedestal = 2.73 m, Distance = 2.73 m (C) Height of pedestal = 2.73 m, Distance = 2.73 m (D) Height of pedestal = 2.73 m, Distance = 2.73 m

ANSWER: (A) Height of pedestal = 2.73 m, Distance = 2.73 m

YEAR: 2026

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: Jigyasa is fly fishing sitting at height 3.1 m above water. Fishing rod is $2\sqrt{2}$ m long and makes 45° with horizontal. The fly at the end of the string resting on water is

5 m away from the base of the platform. The string makes 60° with horizontal. (i) Find height of top of rod above water. (ii) How far is the fly from the point directly below the top of the rod? (iii)(a) Find length of string when fully taut. (Use $\sqrt{3} = 1.7$)

OPTIONS: (A) Height = 5.1 m, Distance = 3 m, String length = 6 m (B) Height = 5.1 m, Distance = 3 m, String length = 6 m (C) Height = 5.1 m, Distance = 3 m, String length = 6 m (D) Height = 5.1 m, Distance = 3 m, String length = 6 m

ANSWER: (A) Height = 5.1 m, Distance = 3 m, String length = 6 m

YEAR: 2026

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: Jigyasa is fly fishing sitting at height 3.1 m above water. Fishing rod is $2\sqrt{2}$ m long and makes 45° with horizontal. The fly at the end of the string resting on water is 5 m away from the base of the platform. The string makes 60° with horizontal. (i) Find height of top of rod above water. (ii) How far is the fly from the point directly below the top of the rod? (iii)(b) If the fish pulled the string making 30° angle with horizontal, find length of string.

OPTIONS: (A) Height = 5.1 m, Distance = 3 m, New string length = 10.2 m (B) Height = 5.1 m, Distance = 3 m, New string length = 10.2 m (C) Height = 5.1 m, Distance = 3 m, New string length = 10.2 m (D) Height = 5.1 m, Distance = 3 m, New string length = 10.2 m

ANSWER: (A) Height = 5.1 m, Distance = 3 m, New string length = 10.2 m

YEAR: 2026

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: Two motorboats A and B are waiting at opposite banks of a river. From a point P on a bridge 20 m above the river, the angles of depression of the boats are 30° and 45° respectively. Both boats leave at speeds 10 m/s and 5 m/s respectively. (i) Find distance travelled by boat A to reach point D vertically below P. (Use $\sqrt{3} = 1.73$) (ii) Find width of river. (iii)(a) Which boat reaches D first and how much earlier?

OPTIONS: (A) Boat A distance = 34.6 m, River width = 54.6 m, Boat A reaches 1.46 s earlier (B) Boat A distance = 34.6 m, River width = 54.6 m, Boat A reaches 1.46 s earlier (C) Boat A distance = 34.6 m, River width = 54.6 m, Boat A reaches 1.46 s earlier (D) Boat A distance = 34.6 m, River width = 54.6 m, Boat A reaches 1.46 s earlier

ANSWER: (A) Boat A distance = 34.6 m, River width = 54.6 m, Boat A reaches 1.46 s earlier

YEAR: 2026

CHAPTER: 10 - HEIGHTS AND DISTANCES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: Two motorboats A and B are waiting at opposite banks of a river. From a point P on a bridge 20 m above the river, the angles of depression of the boats are 30° and 45° respectively. Both boats leave at speeds 10 m/s and 5 m/s respectively. (i) Find distance travelled by boat A to reach point D vertically below P. (Use $\sqrt{3} = 1.73$) (ii) Find width of river. (iii)(b) What is the distance between the two boats after 3 seconds?

OPTIONS: (A) Boat A distance = 34.6 m, River width = 54.6 m, Distance after 3s = 15 m (B) Boat A distance = 34.6 m, River width = 54.6 m, Distance after 3s = 15 m (C) Boat A distance = 34.6 m, River width = 54.6 m, Distance after 3s = 15 m (D) Boat A distance = 34.6 m, River width = 54.6 m, Distance after 3s = 15 m

ANSWER: (A) Boat A distance = 34.6 m, River width = 54.6 m, Distance after 3s = 15 m

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CHAPTER: 11 - AREAS RELATED TO CIRCLES

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YEAR: 2026

CHAPTER: 11 - AREAS RELATED TO CIRCLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The ratio of the area of a quadrant of a circle to the area of the same circle is:

OPTIONS: (A) 1:2 (B) 2:1 (C) 1:4 (D) 4:1

ANSWER: (C) 1:4

YEAR: 2026

CHAPTER: 11 - AREAS RELATED TO CIRCLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The perimeter of a quadrant of a circle of radius r is:

OPTIONS: (A) $\frac{1}{4} \pi r^2$ (B) $\frac{1}{4} \pi r^2 + 2r$ (C) $\frac{\pi r}{2}$ (D) $\frac{\pi r}{2} + 2r$

ANSWER: (D) $\frac{\pi r}{2} + 2r$

YEAR: 2026

CHAPTER: 11 - AREAS RELATED TO CIRCLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The perimeter of a quadrant of a circle of circumference 22 cm is:

OPTIONS: (A) 29 cm (B) 22 cm (C) 12.5 cm (D) 5.5 cm

ANSWER: (C) 12.5 cm

YEAR: 2026

CHAPTER: 11 - AREAS RELATED TO CIRCLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The angle of the sector of a circle whose area is one-eighth of the area of the circle is:

OPTIONS: (A) 22.5° (B) 45° (C) 60° (D) 90°

ANSWER: (B) 45°

YEAR: 2026

CHAPTER: 11 - AREAS RELATED TO CIRCLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Chord AB subtends an angle of 40° at the centre of the circle of radius 9 cm. The length of arc AB is:

OPTIONS: (A) $22/7$ cm (B) $198/7$ cm (C) $44/7$ cm (D) $54/7$ cm

ANSWER: (C) $44/7$ cm

YEAR: 2026

CHAPTER: 11 - AREAS RELATED TO CIRCLES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: An arc of length 11 cm subtends an angle of 105° at the centre of the circle. The radius of the circle is:

OPTIONS: (A) 8 cm (B) $4\sqrt{3}$ cm (C) 6 cm (D) 7 cm

ANSWER: (C) 6 cm

YEAR: 2026

CHAPTER: 11 - AREAS RELATED TO CIRCLES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: The area of a smaller circle is equal to the area of a sector of a larger circle with central angle 120° . The radii of the smaller and larger circles are r and R respectively. Find $r:R$.

OPTIONS: (A) $1:\sqrt{3}$ (B) $1:\sqrt{3}$ (C) $1:\sqrt{3}$ (D) $1:\sqrt{3}$

ANSWER: (A) $1:\sqrt{3}$

YEAR: 2026

CHAPTER: 11 - AREAS RELATED TO CIRCLES

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A chord of a circle of radius 14 cm subtends a right angle at the centre. Find the area of the corresponding (i) minor segment (ii) major segment.

OPTIONS: (A) Minor = 56 cm^2 , Major = 560 cm^2 (B) Minor = 56 cm^2 , Major = 560 cm^2 (C) Minor = 56 cm^2 , Major = 560 cm^2 (D) Minor = 56 cm^2 , Major = 560 cm^2

ANSWER: (A) Minor = 56 cm^2 , Major = 560 cm^2

YEAR: 2026

CHAPTER: 11 - AREAS RELATED TO CIRCLES

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Chord AB of a circle subtends an angle of 120° at the centre O of the circle. Find the length of arc AB, if radius of the circle is 21 cm.

OPTIONS: (A) 44 cm (B) 44 cm (C) 44 cm (D) 44 cm

ANSWER: (A) 44 cm

YEAR: 2026

CHAPTER: 11 - AREAS RELATED TO CIRCLES

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A circle of radius 12 cm is divided into 11 identical sectors. Find the total area of 7 such sectors.

OPTIONS: (A) $288\pi/11 \text{ cm}^2$ (B) $288\pi/11 \text{ cm}^2$ (C) $288\pi/11 \text{ cm}^2$ (D) $288\pi/11 \text{ cm}^2$

ANSWER: (A) $288\pi/11 \text{ cm}^2$

YEAR: 2026

CHAPTER: 11 - AREAS RELATED TO CIRCLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: (a) ABCD is a square. With centres A, B, C and D, four quadrants (each touching two of the remaining three) are drawn inside the square ABCD. If the area of the shaded region is 42 cm^2 , find the side of the square ABCD. Also, find the perimeter of the shaded region.

OPTIONS: (A) Side = 14 cm, Perimeter = 44 cm (B) Side = 14 cm, Perimeter = 44 cm (C) Side = 14 cm, Perimeter = 44 cm (D) Side = 14 cm, Perimeter = 44 cm

ANSWER: (A) Side = 14 cm, Perimeter = 44 cm

YEAR: 2026

CHAPTER: 11 - AREAS RELATED TO CIRCLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: (b) A rectangle ABCD with diagonal 14 cm is inscribed in a circle with centre O. If the area of the shaded portion is expressed as $a + b\sqrt{3}$, find the values of a and b. Also, find the perimeter of the sector OABO.

OPTIONS: (A) $a = 154/3$, $b = -49/2$, Perimeter = $86/3$ cm (B) $a = 154/3$, $b = -49/2$, Perimeter = $86/3$ cm (C) $a = 154/3$, $b = -49/2$, Perimeter = $86/3$ cm (D) $a = 154/3$, $b = -49/2$, Perimeter = $86/3$ cm

ANSWER: (A) $a = 154/3$, $b = -49/2$, Perimeter = $86/3$ cm

YEAR: 2026

CHAPTER: 11 - AREAS RELATED TO CIRCLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: In the adjoining figure, $\triangle OAB$ is an equilateral triangle and the area of the shaded region is 750π cm². Find the perimeter of the shaded region.

OPTIONS: (A) $50\pi + 60 = 217.14$ cm (B) $50\pi + 60 = 217.14$ cm (C) $50\pi + 60 = 217.14$ cm (D) $50\pi + 60 = 217.14$ cm

ANSWER: (A) $50\pi + 60 = 217.14$ cm

YEAR: 2026

CHAPTER: 11 - AREAS RELATED TO CIRCLES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: O and O' are the centres of the circles of radius r as shown in figures (i) and (ii) respectively. Find the ratio of area of shaded region in figure (i) to that of area of shaded region in figure (ii).

OPTIONS: (A) $(4\pi - 3\sqrt{3}) : (2\pi - 3\sqrt{3})$ (B) $(4\pi - 3\sqrt{3}) : (2\pi - 3\sqrt{3})$ (C) $(4\pi - 3\sqrt{3}) : (2\pi - 3\sqrt{3})$ (D) $(4\pi - 3\sqrt{3}) : (2\pi - 3\sqrt{3})$

ANSWER: (A) $(4\pi - 3\sqrt{3}) : (2\pi - 3\sqrt{3})$

YEAR: 2026

CHAPTER: 11 - AREAS RELATED TO CIRCLES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A farmer has a semicircular field with centre O and diameter 120 m. In segment ACPA he grows rice. In triangular region APB he grows vegetables. In segment BDPB cows graze. $\angle PAO = 30^\circ$ and $\angle POB = 60^\circ$. (i) Find radius and $\angle ABP$. (ii) Find area of sector AOP. (iii)(a) Find area of region where cows graze. (Use $\pi = 3.14$)

OPTIONS: (A) Radius = 60 m, $\angle ABP = 60^\circ$, Area = $1884 - 900\sqrt{3} \text{ m}^2$ (B) Radius = 60 m, $\angle ABP = 60^\circ$, Area = $1884 - 900\sqrt{3} \text{ m}^2$ (C) Radius = 60 m, $\angle ABP = 60^\circ$, Area = $1884 - 900\sqrt{3} \text{ m}^2$ (D) Radius = 60 m, $\angle ABP = 60^\circ$, Area = $1884 - 900\sqrt{3} \text{ m}^2$

ANSWER: (A) Radius = 60 m, $\angle ABP = 60^\circ$, Area = $1884 - 900\sqrt{3} \text{ m}^2$

YEAR: 2026

CHAPTER: 11 - AREAS RELATED TO CIRCLES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A farmer has a semicircular field with centre O and diameter 120 m. In segment ACPA he grows rice. In triangular region APB he grows vegetables. In segment BDPB cows graze. $\angle PAO = 30^\circ$ and $\angle POB = 60^\circ$. (i) Find radius and $\angle ABP$. (ii) Find area of sector AOP. (iii)(b) Find area of region where vegetables are grown.

OPTIONS: (A) Area = $1800\sqrt{3} \text{ m}^2$ (B) Area = $1800\sqrt{3} \text{ m}^2$ (C) Area = $1800\sqrt{3} \text{ m}^2$ (D) Area = $1800\sqrt{3} \text{ m}^2$

ANSWER: (A) Area = $1800\sqrt{3} \text{ m}^2$

YEAR: 2026

CHAPTER: 11 - AREAS RELATED TO CIRCLES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: A circular sheet of radius 70 cm, a quadrant is cut. Find the area of the remaining sheet.

OPTIONS: (A) $3675\pi \text{ cm}^2$ (B) $3675\pi \text{ cm}^2$ (C) $3675\pi \text{ cm}^2$ (D) $3675\pi \text{ cm}^2$

ANSWER: (A) $3675\pi \text{ cm}^2$

YEAR: 2026

CHAPTER: 11 - AREAS RELATED TO CIRCLES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: In the given figure, O is the centre of a circle of radius 7 cm. AB is a chord of the circle. Find the perimeter of the shaded region.

OPTIONS: (A) 44 cm (B) 44 cm (C) 44 cm (D) 44 cm

ANSWER: (A) 44 cm

YEAR: 2026

CHAPTER: 11 - AREAS RELATED TO CIRCLES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: In the given figure, OABC is a quadrant of a circle with centre O and radius 8 cm. If OD = 6 cm, find the perimeter of the shaded region.

OPTIONS: (A) $2\pi + 8$ cm (B) $2\pi + 8$ cm (C) $2\pi + 8$ cm (D) $2\pi + 8$ cm

ANSWER: (A) $2\pi + 8$ cm

YEAR: 2026

CHAPTER: 11 - AREAS RELATED TO CIRCLES

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: In the given figure, OPRQ is a quadrant of a circle with centre O and radius 10 cm. If OP = 10 cm and OS = 6 cm, find the area of the shaded region. (Use $\pi = 3.14$)

OPTIONS: (A) $78.5 - 24 = 54.5 \text{ cm}^2$ (B) $78.5 - 24 = 54.5 \text{ cm}^2$ (C) $78.5 - 24 = 54.5 \text{ cm}^2$ (D) $78.5 - 24 = 54.5 \text{ cm}^2$

ANSWER: (A) $78.5 - 24 = 54.5 \text{ cm}^2$

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CHAPTER: 12 - SURFACE AREAS AND VOLUMES

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YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: For which of the following solids is the lateral/curved surface area and total surface area the same?

OPTIONS: (A) Cube (B) Cuboid (C) Hemisphere (D) Sphere

ANSWER: (D) Sphere

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The radius of the largest sphere that can be carved out from a solid cube of side 14 cm is:

OPTIONS: (A) 7 cm (B) 14 cm (C) $14\sqrt{2}$ cm (D) $7\sqrt{2}$ cm

ANSWER: (A) 7 cm

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Two wooden solid hemispheres of same radii r , are joined at a point. The total surface area of the object is:

OPTIONS: (A) $4\pi r^2$ (B) $6\pi r^2$ (C) $3\pi r^2$ (D) $5\pi r^2$

ANSWER: (B) $6\pi r^2$

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A cylinder of radius r is surmounted on a hemisphere of same radius. If total height of the object is 13 cm, then its inner surface area is:

OPTIONS: (A) $2\pi r(r+13)$ (B) $13\pi r$ (C) $2\pi(13+r)^2$ (D) $26\pi r$

ANSWER: (D) $26\pi r$

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The total surface area of a solid cone of radius 7 cm and slant height 25 cm is:

OPTIONS: (A) 724 cm^2 (B) 704 cm^2 (C) 550 cm^2 (D) 616 cm^2

ANSWER: (B) 704 cm^2

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A cone with slant height 10 cm and radius 6 cm is surmounted on a hemisphere of same radius. The height of the toy is:

OPTIONS: (A) 12 cm (B) 16 cm (C) 14 cm (D) 10 cm

ANSWER: (C) 14 cm

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The largest possible right circular cone is carved out of a solid hemisphere of radius r . The slant height of the cone is:

OPTIONS: (A) r (B) $2r$ (C) $\sqrt{2}r$ (D) $\sqrt{3}r$

ANSWER: (C) $\sqrt{2}r$

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A cone and a cylinder have the same base radius and volume. The (height of cone : height of cylinder) is:

OPTIONS: (A) 1:1 (B) 3:1 (C) 1:3 (D) 3:2

ANSWER: (B) 3:1

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A cone and cylinder have same height and same radius. The volume of the cone and the volume of the cylinder are in the ratio:

OPTIONS: (A) 1:1 (B) 1:3 (C) 3:1 (D) 1:2

ANSWER: (B) 1:3

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A cone of height h and radius r is surmounted on a solid cylinder of same dimensions. The total surface area of the entire solid will be:

OPTIONS: (A) $2\pi rh + \pi r\sqrt{h^2+r^2}$ (B) $2\pi rh + \pi r^2 + \pi r\sqrt{h^2+r^2}$ (C) $2\pi rh + 2\pi r^2 + \pi r\sqrt{h^2+r^2}$ (D) $2\pi rh + \pi r\sqrt{h^2+r^2} - \pi r^2$

ANSWER: (B) $2\pi rh + \pi r^2 + \pi r\sqrt{h^2+r^2}$

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A toy is in the form of a cone surmounted on a hemisphere. The cone and hemisphere have the same radii. The height of the conical part is equal to the diameter of its base. If the radius of the conical part is 5 cm, find the volume of the toy.

OPTIONS: (A) $500\pi/3 \text{ cm}^3$ (B) $500\pi/3 \text{ cm}^3$ (C) $500\pi/3 \text{ cm}^3$ (D) $500\pi/3 \text{ cm}^3$

ANSWER: (A) $500\pi/3 \text{ cm}^3$

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A cubical block is surmounted by a hemisphere of radius 3.5 cm. What is the smallest possible length of the edge of the cube so that the hemisphere can totally lie on the cube? Find the total surface area of the solid so formed.

OPTIONS: (A) Edge = 7 cm, TSA = 332.5 cm^2 (B) Edge = 7 cm, TSA = 332.5 cm^2 (C) Edge = 7 cm, TSA = 332.5 cm^2 (D) Edge = 7 cm, TSA = 332.5 cm^2

ANSWER: (A) Edge = 7 cm, TSA = 332.5 cm^2

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A wooden article was made by scooping out a hemisphere (of same diameter) from one end of a solid cylinder. If the height of the cylinder is 10 cm and diameter is 14 cm, find the total surface area of the remaining wooden article. (Use $\pi = \frac{22}{7}$)

OPTIONS: (A) $2.94\pi \text{ m}^2$ (B) $2.94\pi \text{ m}^2$ (C) $2.94\pi \text{ m}^2$ (D) $2.94\pi \text{ m}^2$

ANSWER: (A) $2.94\pi \text{ m}^2$

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A solid wooden cylinder of height 10 cm and radius 14 cm has a cylinder of radius 7 cm and height 5 cm scooped out to form a cavity. Find the total surface area of the remaining solid.

OPTIONS: (A) 2170 cm^2 (B) 2170 cm^2 (C) 2170 cm^2 (D) 2170 cm^2

ANSWER: (A) 2170 cm^2

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A perfume bottle is in the form of a cylinder but the bottom has a hemispherical raised portion to reduce capacity. Inner diameter is 5 cm and height is 10 cm. Find capacity in mL. (Use $\pi = 3.14$, $1 \text{ cm}^3 = 1 \text{ mL}$)

OPTIONS: (A) 163.54 mL (B) 163.54 mL (C) 163.54 mL (D) 163.54 mL

ANSWER: (A) 163.54 mL

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A hemispherical depression is scooped out from the top face of a wooden cubical block of side 14 cm. If the diameter of the hemisphere is equal to the side of the cube, find the total surface area of the remaining solid block. (Use $\pi = 22/7$)

OPTIONS: (A) 1330 cm^2 (B) 1330 cm^2 (C) 1330 cm^2 (D) 1330 cm^2

ANSWER: (A) 1330 cm^2

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A pen stand is made up of wood in the shape of a cuboid with four conical depressions and cylindrical extensions. Dimensions of cuboid: $15 \text{ cm} \times 10 \text{ cm} \times 3.5 \text{ cm}$. Radius of each conical depression = 0.5 cm, depth = 1.4 cm. Radius and height of each cylindrical extension = 0.5 cm and 4.9 cm. (i) Find volume of wood in entire stand. (ii) Find inner surface area of a conical depression along with cylindrical extension. (Use $\sqrt{221} = 14.8$)

OPTIONS: (A) Volume = 523.53 cm^3 , Surface area = 17.72 cm^2 (B) Volume = 523.53 cm^3 , Surface area = 17.72 cm^2 (C) Volume = 523.53 cm^3 , Surface area = 17.72 cm^2 (D) Volume = 523.53 cm^3 , Surface area = 17.72 cm^2

ANSWER: (A) Volume = 523.53 cm^3 , Surface area = 17.72 cm^2

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A solid is in the shape of a cylinder with a solid hemisphere attached to one of its ends. Let the height and diameter of the cylinder be same (say H). Prove that the volume of the solid is same as the volume of a cone of height H and diameter $2H$. Also find the total surface area of the solid in terms of H .

OPTIONS: (A) Volume = $\pi H^3/3$, TSA = $7\pi H^2/4$ (B) Volume = $\pi H^3/3$, TSA = $7\pi H^2/4$ (C) Volume = $\pi H^3/3$, TSA = $7\pi H^2/4$ (D) Volume = $\pi H^3/3$, TSA = $7\pi H^2/4$

ANSWER: (A) Volume = $\pi H^3/3$, TSA = $7\pi H^2/4$

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A spherical glass vessel has a cylindrical neck which is 7 cm long and 2 cm in diameter. The diameter of the spherical part is 14 cm. Find the capacity of the entire glass vessel. (Use $\pi = 22/7$)

OPTIONS: (A) 1437.33 cm³ (B) 1437.33 cm³ (C) 1437.33 cm³ (D) 1437.33 cm³

ANSWER: (A) 1437.33 cm³

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: As a part of school project, Mishika and Sahaj created a bird-bath from a cylindrical log by scooping out a hemispherical depression from one end. Cylinder has length 2 m, out of which 0.6 m is in earth, and diameter is 1.4 m. (i) Write radius of hemispherical depression. (ii) Find volume of water that can be filled in the hemispherical depression in terms of π . (iii)(a) Find total surface area of log above the ground after making the bird-bath.

OPTIONS: (A) $r = 0.7$ m, Volume = 0.228π m³, TSA = 2.94π m² (B) $r = 0.7$ m, Volume = 0.228π m³, TSA = 2.94π m² (C) $r = 0.7$ m, Volume = 0.228π m³, TSA = 2.94π m² (D) $r = 0.7$ m, Volume = 0.228π m³, TSA = 2.94π m²

ANSWER: (A) $r = 0.7$ m, Volume = 0.228π m³, TSA = 2.94π m²

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: As a part of school project, Mishika and Sahaj created a bird-bath from a cylindrical log by scooping out a hemispherical depression from one end. Cylinder has length 2 m, out of which 0.6 m is in earth, and diameter is 1.4 m. (i) Write radius of hemispherical depression. (ii) Find volume of water that can be filled in the hemispherical depression in terms of π . (iii)(b) Compute the volume of log above the ground after making the bird-bath.

OPTIONS: (A) $r = 0.7$ m, Volume = $0.228\pi \text{ m}^3$, Volume of log = $0.457\pi \text{ m}^3$ (B) $r = 0.7$ m, Volume = $0.228\pi \text{ m}^3$, Volume of log = $0.457\pi \text{ m}^3$ (C) $r = 0.7$ m, Volume = $0.228\pi \text{ m}^3$, Volume of log = $0.457\pi \text{ m}^3$ (D) $r = 0.7$ m, Volume = $0.228\pi \text{ m}^3$, Volume of log = $0.457\pi \text{ m}^3$

ANSWER: (A) $r = 0.7$ m, Volume = $0.228\pi \text{ m}^3$, Volume of log = $0.457\pi \text{ m}^3$

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: A necklace is made up of wooden beads. Each bead is in the form of a sphere of diameter 4.2 mm. A cylinder is hollowed out from each bead. If the radius of the cylinder is 1 mm, find the volume of wood left in each bead.

OPTIONS: (A) $38.78 - 6.28 = 32.5 \text{ mm}^3$ (B) $38.78 - 6.28 = 32.5 \text{ mm}^3$ (C) $38.78 - 6.28 = 32.5 \text{ mm}^3$ (D) $38.78 - 6.28 = 32.5 \text{ mm}^3$

ANSWER: (A) $38.78 - 6.28 = 32.5 \text{ mm}^3$

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: For a cricket tournament, a special trophy is designed. The height and diameter of the cylindrical part are 14 cm and 6 cm respectively. The diameter of the spherical ball on the top is 7 cm. (i) Find total height of trophy excluding wooden part. (ii) Find difference between radius of sphere and cylinder. (iii)(a) If cylindrical and spherical parts are separated and gold plated overall, find total surface area to be gold plated.

OPTIONS: (A) Height = 21 cm, Difference = 0.5 cm, TSA = 474.57 cm^2 (B) Height = 21 cm, Difference = 0.5 cm, TSA = 474.57 cm^2 (C) Height = 21 cm, Difference = 0.5 cm, TSA = 474.57 cm^2 (D) Height = 21 cm, Difference = 0.5 cm, TSA = 474.57 cm^2

ANSWER: (A) Height = 21 cm, Difference = 0.5 cm, TSA = 474.57 cm^2

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: For a cricket tournament, a special trophy is designed. The height and diameter of the cylindrical part are 14 cm and 6 cm respectively. The diameter of the spherical ball on the top is 7 cm. (i) Find total height of trophy excluding wooden part. (ii) Find difference between radius of sphere and cylinder. (iii)(b) Find volume of metal used in making the trophy, assuming metal is completely filled in it.

OPTIONS: (A) Height = 21 cm, Difference = 0.5 cm, Volume = 575.67 cm^3 (B) Height = 21 cm, Difference = 0.5 cm, Volume = 575.67 cm^3 (C) Height = 21 cm, Difference = 0.5 cm, Volume = 575.67 cm^3 (D) Height = 21 cm, Difference = 0.5 cm, Volume = 575.67 cm^3

ANSWER: (A) Height = 21 cm, Difference = 0.5 cm, Volume = 575.67 cm^3

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: 'Gilli Danda' is a traditional game. Danda is cylindrical with diameter 4 cm and length 42 cm. Gilli has a cylindrical middle with identical conical ends. Radius of conical ends = 1.5 cm, length = 2.8 cm. Length of cylindrical part = 7 cm. (i) Find volume of wood in both conical parts. (ii) Find volume of wood in cylindrical part. (iii)(a) A cylindrical log of radius 1.5 cm and length 14 cm is used to make Gilli. Find volume of wood scrapped.

OPTIONS: (A) Conical = 13.2 cm^3 , Cylindrical = 49.5 cm^3 , Scrapped = 36.3 cm^3 (B) Conical = 13.2 cm^3 , Cylindrical = 49.5 cm^3 , Scrapped = 36.3 cm^3 (C) Conical = 13.2 cm^3 , Cylindrical = 49.5 cm^3 , Scrapped = 36.3 cm^3 (D) Conical = 13.2 cm^3 , Cylindrical = 49.5 cm^3 , Scrapped = 36.3 cm^3

ANSWER: (A) Conical = 13.2 cm^3 , Cylindrical = 49.5 cm^3 , Scrapped = 36.3 cm^3

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: 'Gilli Danda' is a traditional game. Danda is cylindrical with diameter 4 cm and length 42 cm. Gilli has a cylindrical middle with identical conical ends. Radius of conical ends = 1.5 cm, length = 2.8 cm. Length of cylindrical part = 7 cm. (i) Find volume of wood in both conical parts. (ii) Find volume of wood in cylindrical part. (iii)(b) Find total surface area of Danda.

OPTIONS: (A) TSA = 553.14 cm² (B) TSA = 553.14 cm² (C) TSA = 553.14 cm² (D) TSA = 553.14 cm²

ANSWER: (A) TSA = 553.14 cm²

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: There are many varieties of mushrooms. Amanita muscaria has an upper part like a red cap (hemispherical) and lower part like a white stem (cylindrical).

Hemispherical cap radius = 3 cm. Cylindrical stem is 2 cm high with diameter 1.4 cm. (i) What is total height of mushroom? (ii) Find volume of stem. (iii)(a) Determine volume of 7 such mushrooms.

OPTIONS: (A) Height = 5 cm, Volume of stem = 3.08 cm³, Volume of 7 = 417.56 cm³ (B) Height = 5 cm, Volume of stem = 3.08 cm³, Volume of 7 = 417.56 cm³ (C) Height = 5 cm, Volume of stem = 3.08 cm³, Volume of 7 = 417.56 cm³ (D) Height = 5 cm, Volume of stem = 3.08 cm³, Volume of 7 = 417.56 cm³

ANSWER: (A) Height = 5 cm, Volume of stem = 3.08 cm³, Volume of 7 = 417.56 cm³

YEAR: 2026

CHAPTER: 12 - SURFACE AREAS AND VOLUMES

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: There are many varieties of mushrooms. Amanita muscaria has an upper part like a red cap (hemispherical) and lower part like a white stem (cylindrical).

Hemispherical cap radius = 3 cm. Cylindrical stem is 2 cm high with diameter 1.4 cm. (i) What is total height of mushroom? (ii) Find volume of stem. (iii)(b) Find total surface area of 7 such mushrooms.

OPTIONS: (A) Height = 5 cm, Volume of stem = 3.08 cm³, TSA = 655.6 cm² (B) Height = 5 cm, Volume of stem = 3.08 cm³, TSA = 655.6 cm² (C) Height = 5 cm, Volume of stem = 3.08 cm³, TSA = 655.6 cm² (D) Height = 5 cm, Volume of stem = 3.08 cm³, TSA = 655.6 cm²

ANSWER: (A) Height = 5 cm, Volume of stem = 3.08 cm^3 , TSA = 655.6 cm^2

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CHAPTER: 13 - STATISTICS

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YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The class mark of the median class of the following data is: [Table: Class
10-25 (2), 25-40 (3), 40-55 (7), 55-70 (6), 70-85 (6), 85-100 (6)]

OPTIONS: (A) 40 (B) 55 (C) 47.5 (D) 62.5

ANSWER: (C) 47.5

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The upper limit of the median class of the following data is: [Table: Class
0-10 (3), 10-20 (5), 20-30 (7), 30-40 (9), 40-50 (11)]

OPTIONS: (A) 10 (B) 20 (C) 30 (D) 40

ANSWER: (D) 40

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The lower limit of the modal class of the following distribution is: [Table:
3000-4000 (5), 4000-5000 (10), 5000-6000 (9), 6000-7000 (8)]

OPTIONS: (A) 3000 (B) 4000 (C) 5000 (D) 6000

ANSWER: (B) 4000

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The lower limit of the modal class of the following data is: [Table: 0-10 (5), 10-20 (3), 20-30 (4), 30-40 (8), 40-50 (3)]

OPTIONS: (A) 10 (B) 20 (C) 30 (D) 40

ANSWER: (C) 30

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the formula of mode given by $\text{mode} = l + \frac{(f_1 - f_0)}{(2f_1 - f_0 - f_2)} \times h$, f_1 denotes the:

OPTIONS: (A) frequency of the modal class (B) frequency of class preceding modal class (C) frequency of class succeeding modal class (D) cumulative frequency of modal class

ANSWER: (A) frequency of the modal class

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: For a distribution, if mean = median = a, then its mode is:

OPTIONS: (A) 3a (B) 2a (C) a (D) 0

ANSWER: (C) a

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If for a frequency distribution, the mean is $\frac{3}{4}$ times the median, then mode is:

OPTIONS: (A) equal to median (B) $\frac{3}{2}$ times the median (C) equal to mean (D) 3 times the mean

ANSWER: (B) $\frac{3}{2}$ times the median

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If for a data, median is 5 and mode is 4, then mean is equal to:

OPTIONS: (A) 7 (B) 11 (C) $11/2$ (D) $14/3$

ANSWER: (C) $11/2$

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: For a distribution, if mean = 15 and mode = 12, then its median is:

OPTIONS: (A) 12 (B) 13 (C) 14 (D) 15

ANSWER: (C) 14

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following is affected by the extreme values in a given data?

OPTIONS: (A) Mean only (B) Median only (C) Mode only (D) Mean and Mode both

ANSWER: (A) Mean only

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Let 'a' be the assumed mean and 'h' be the class size for a grouped data.

Which of the following is not the correct formula to find the mean of the grouped data?

OPTIONS: (A) $\bar{x} = \sum f_i x_i / \sum f_i$ (B) $\bar{x} = a + \sum f_i d_i / \sum f_i$, where $d_i = x_i - a$ (C) $\bar{x} = a + \sum f_i u_i / \sum f_i$, where $u_i = (x_i - a)/h$ (D) $\bar{x} = a + \sum f_i y_i / \sum f_i \times h$, where $y_i = (x_i - a)/h$

ANSWER: (C) $\bar{x} = a + \sum f_i u_i / \sum f_i$, where $u_i = (x_i - a)/h$

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: A box consists of 60 wall clocks, out of which 40 are good, 15 have minor defects and the remaining are broken. Find probability that (i) the box will be rejected (ii) the clock has minor defect.

OPTIONS: (A) (i) $\frac{1}{12}$, (ii) $\frac{1}{4}$ (B) (i) $\frac{1}{12}$, (ii) $\frac{1}{4}$ (C) (i) $\frac{1}{12}$, (ii) $\frac{1}{4}$ (D) (i) $\frac{1}{12}$, (ii) $\frac{1}{4}$

ANSWER: (A) (i) $\frac{1}{12}$, (ii) $\frac{1}{4}$

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The following data gives the information on the observed lifetime (in hours) of 200 electrical components. Find the mean lifetime. [Table: 0-20 (10), 20-40 (35), 40-60 (50), 60-80 (60), 80-100 (30), 100-120 (15)]

OPTIONS: (A) 61 hours (B) 61 hours (C) 61 hours (D) 61 hours

ANSWER: (A) 61 hours

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The following frequency distribution gives the monthly consumption of electricity of 68 consumers. Find the mean consumption. [Table: 50-100 (4), 100-150 (5), 150-200 (13), 200-250 (20), 250-300 (14), 300-350 (8), 350-400 (4)]

OPTIONS: (A) 211.76 units (B) 211.76 units (C) 211.76 units (D) 211.76 units

ANSWER: (A) 211.76 units

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Find the median and the mode for the following data: [Table: 0-20 (15), 20-40 (18), 40-60 (21), 60-80 (29), 80-100 (17)]

OPTIONS: (A) Median = 56.19, Mode = 68 (B) Median = 56.19, Mode = 68 (C) Median = 56.19, Mode = 68 (D) Median = 56.19, Mode = 68

ANSWER: (A) Median = 56.19, Mode = 68

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The median of the following data is 525. If the sum of all frequencies is 100, find p and q. [Table: 0-100 (2), 100-200 (p), 200-300 (9), 300-400 (12), 400-500 (17), 500-600 (20), 600-700 (15), 700-800 (9), 800-900 (q), 900-1000 (4)]

OPTIONS: (A) $p = 5, q = 7$ (B) $p = 5, q = 7$ (C) $p = 5, q = 7$ (D) $p = 5, q = 7$

ANSWER: (A) $p = 5, q = 7$

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The median of the following distribution is 137. Find x and y. [Table: 65-85 (4), 85-105 (5), 105-125 (x), 125-145 (20), 145-165 (14), 165-185 (y), 185-205 (4), Total = 68]

OPTIONS: (A) $x = 13, y = 8$ (B) $x = 13, y = 8$ (C) $x = 13, y = 8$ (D) $x = 13, y = 8$

ANSWER: (A) $x = 13, y = 8$

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A life insurance agent found the following data for the distribution of 100 policy holders on the basis of their ages. Find the median age. [Table: 15-20 (2), 20-25 (4), 25-30 (18), 30-35 (21), 35-40 (33), 40-45 (11), 45-50 (3), 50-55 (6), 55-60 (2)]

OPTIONS: (A) 35.76 years (B) 35.76 years (C) 35.76 years (D) 35.76 years

ANSWER: (A) 35.76 years

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The following table shows the weight (in kg) of 40 students. Find the mean weight and median weight. [Table: 40-45 (3), 45-50 (4), 50-55 (10), 55-60 (8), 60-65 (8), 65-70 (4), 70-75 (3)]

OPTIONS: (A) Mean = 57.25, Median = 56.87 (B) Mean = 57.25, Median = 56.87 (C) Mean = 57.25, Median = 56.87 (D) Mean = 57.25, Median = 56.87

ANSWER: (A) Mean = 57.25, Median = 56.87

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A life insurance agent insured 100 people of different ages. Calculate the mean age and median age. [Table: 15-20 (2), 20-25 (4), 25-30 (18), 30-35 (21), 35-40 (33), 40-45 (11), 45-50 (3), 50-55 (6), 55-60 (2)]

OPTIONS: (A) Mean = 35.8, Median = 35.76 (B) Mean = 35.8, Median = 35.76 (C) Mean = 35.8, Median = 35.76 (D) Mean = 35.8, Median = 35.76

ANSWER: (A) Mean = 35.8, Median = 35.76

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The following frequency distribution shows the monthly consumption of electricity of 60 families. Find the mean and median. [Table: 200-300 (3), 300-400 (4), 400-500 (12), 500-600 (18), 600-700 (13), 700-800 (7), 800-900 (3)]

OPTIONS: (A) Mean = 550, Median = 550 (B) Mean = 550, Median = 550 (C) Mean = 550, Median = 550 (D) Mean = 550, Median = 550

ANSWER: (A) Mean = 550, Median = 550

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The following distribution shows the weekly pocket allowance (in ₹) of some children. The mean pocket allowance is ₹180. Find f. Hence find the mode. [Table:

110-130 (7), 130-150 (6), 150-170 (9), 170-190 (13), 190-210 (f), 210-230 (5), 230-250 (4)]

OPTIONS: (A) $f = 6$, Mode = 188.33 (B) $f = 6$, Mode = 188.33 (C) $f = 6$, Mode = 188.33 (D) $f = 6$, Mode = 188.33

ANSWER: (A) $f = 6$, Mode = 188.33

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The following table shows the daily expenditure of 25 households. Find the value of x . Hence find the mean daily expenditure. [Table: 500-750 (4), 750-1000 ($2x+1$), 1000-1250 (12), 1250-1500 (x), 1500-1750 (2)]

OPTIONS: (A) $x = 2$, Mean = 1125 (B) $x = 2$, Mean = 1125 (C) $x = 2$, Mean = 1125 (D) $x = 2$, Mean = 1125

ANSWER: (A) $x = 2$, Mean = 1125

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: The lengths of 40 leaves of a plant are measured. Find the median length. [Table: 100-120 (8), 120-140 (9), 140-160 (12), 160-180 (5), 180-200 (6)]

OPTIONS: (A) 145 mm (B) 145 mm (C) 145 mm (D) 145 mm

ANSWER: (A) 145 mm

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: A class teacher has the following absentees record of 30 students. If the mean number of days a student was absent is 12, find x and y . [Table: 0-4 (1), 4-8 (8), 8-12 (x), 12-16 (6), 16-20 (5), 20-24 (y)]

OPTIONS: (A) $x = 7$, $y = 3$ (B) $x = 7$, $y = 3$ (C) $x = 7$, $y = 3$ (D) $x = 7$, $y = 3$

ANSWER: (A) $x = 7$, $y = 3$

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: Mansi collected daily AQI data of her city for a month. AQI Range: 1-100 (3 days), 101-200 (9 days), 201-300 (12 days), 301-400 (4 days), 401-500 (2 days). (i) Convert to continuous frequency distribution. (ii) What is the quality of air in most days? (iii)(a) Find mode of the data.

OPTIONS: (A) Mode = 227.77 (B) Mode = 227.77 (C) Mode = 227.77 (D) Mode = 227.77

ANSWER: (A) Mode = 227.77

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: E (Case Study)

ORIGINAL_MARKS: 4

QUESTION: Mansi collected daily AQI data of her city for a month. AQI Range: 1-100 (3 days), 101-200 (9 days), 201-300 (12 days), 301-400 (4 days), 401-500 (2 days). (i) Convert to continuous frequency distribution. (ii) What is the quality of air in most days? (iii)(b) Find median of the data.

OPTIONS: (A) Median = 225.5 (B) Median = 225.5 (C) Median = 225.5 (D) Median = 225.5

ANSWER: (A) Median = 225.5

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CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Find mean and mode of the following data: [Table: Class 10-20 (5), 20-30 (4), 30-40 (10), 40-50 (13), 50-60 (12), 60-70 (10), 70-80 (6)]

OPTIONS: (A) Mean = 47.83, Mode = 47.5 (B) Mean = 47.83, Mode = 47.5 (C) Mean = 47.83, Mode = 47.5 (D) Mean = 47.83, Mode = 47.5

ANSWER: (A) Mean = 47.83, Mode = 47.5

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CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Find mean and mode of the following data: [Table: Class 0-15 (4), 15-30 (6), 30-45 (8), 45-60 (10), 60-75 (12), 75-90 (7), 90-105 (3)]

OPTIONS: (A) Mean = 53.4, Mode = 64.28 (B) Mean = 53.4, Mode = 64.28 (C) Mean = 53.4, Mode = 64.28 (D) Mean = 53.4, Mode = 64.28

ANSWER: (A) Mean = 53.4, Mode = 64.28

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Find the missing frequencies p and q in the following frequency distribution, when sum of frequencies is 40 and mean is 19. [Table: 0-5 (2), 5-10 (5), 10-15 (6), 15-20 (p), 20-25 (10), 25-30 (q), 30-35 (4)]

OPTIONS: (A) p = 6, q = 7 (B) p = 6, q = 7 (C) p = 6, q = 7 (D) p = 6, q = 7

ANSWER: (A) p = 6, q = 7

YEAR: 2026

CHAPTER: 13 - STATISTICS

ORIGINAL_SECTION: D

ORIGINAL_MARKS: 5

QUESTION: Consider the given data: [Table: Class 0-20 (15), 20-40 (30), 40-60 (50), 60-80 (63), 80-100 (35), 100-120 (32)]. The difference of lower limit of the median class and upper limit of the modal class is:

OPTIONS: (A) 0 (B) 20 (C) 40 (D) 10

ANSWER: (B) 20

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CHAPTER: 14 - PROBABILITY

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CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A bag contains 3 red, 4 white and 7 green balls. A ball is drawn at random. The probability that the ball drawn is not of red colour is:

OPTIONS: (A) $1/11$ (B) $3/14$ (C) $11/14$ (D) $3/11$

ANSWER: (C) $11/14$

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A bag contains some red and some white balls. A ball is drawn at random from the bag. If the probability of getting a red ball is $2/7$, then the probability of getting a white ball is:

OPTIONS: (A) $1/14$ (B) $5/7$ (C) $1/7$ (D) $2/7$

ANSWER: (B) $5/7$

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CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A card is drawn from a well-shuffled deck of 52 playing cards. The probability of getting a queen of spade is:

OPTIONS: (A) $1/26$ (B) $1/52$ (C) 0 (D) $1/4$

ANSWER: (B) $1/52$

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A card is drawn from a well shuffled deck of 52 cards. The probability of getting an ace or a ten is:

OPTIONS: (A) $1/13$ (B) $2/13$ (C) $1/26$ (D) $4/13$

ANSWER: (B) $2/13$

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A card is drawn from a well shuffled deck of 52 cards. The probability that it is not a diamond card is:

OPTIONS: (A) $1/4$ (B) 0 (C) $1/2$ (D) $3/4$

ANSWER: (D) $3/4$

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The probability of getting sum greater than 10, when two dice are rolled together, is:

OPTIONS: (A) $1/9$ (B) $1/18$ (C) $1/12$ (D) 1

ANSWER: (C) $1/12$

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In an experiment of throwing a pair of dice, the probability of not getting a doublet is:

OPTIONS: (A) $1/6$ (B) $5/6$ (C) $1/5$ (D) $1/30$

ANSWER: (B) $5/6$

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: An unbiased die is thrown once. The probability of getting an even prime number greater than 2 is:

OPTIONS: (A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{6}$ (D) 0

ANSWER: (D) 0

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: An unbiased die is thrown once. The probability of getting an odd prime number less than 3 is:

OPTIONS: (A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{6}$ (D) 0

ANSWER: (D) 0

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: An unbiased die is thrown once. The probability of getting an even multiple of 3 is:

OPTIONS: (A) 0 (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $\frac{1}{6}$

ANSWER: (D) $\frac{1}{6}$

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Three coins are tossed together. The probability of getting exactly two tails is:

OPTIONS: (A) $\frac{2}{8}$ (B) $\frac{1}{2}$ (C) $\frac{3}{8}$ (D) 1

ANSWER: (C) $\frac{3}{8}$

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: From a bag containing yellow, red and green balls, the probability of drawing a yellow or red ball is equal to that of drawing a green ball. The probability of drawing a green ball from the bag is:

OPTIONS: (A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) 1 (D) Not possible to find

ANSWER: (B) $\frac{1}{2}$

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A letter is selected from the letters of the word FEBRUARY. The probability that it is a vowel is:

OPTIONS: (A) $\frac{1}{8}$ (B) $\frac{2}{8}$ (C) $\frac{3}{8}$ (D) $\frac{3}{7}$

ANSWER: (C) $\frac{3}{8}$

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In a random experiment of throwing a die, which of the following is a sure event?

OPTIONS: (A) Getting a number between 1 and 6 (B) Getting an odd number < 7 (C) Getting an even number < 7 (D) Getting a natural number < 7

ANSWER: (D) Getting a natural number < 7

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The total number of outcomes in the experiment of simultaneous throw of three dice is:

OPTIONS: (A) 6 (B) 18 (C) 36 (D) 216

ANSWER: (D) 216

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A pair of dice is thrown simultaneously. Let E denote the event that "The sum of numbers obtained on both dice is at least 9." The number of outcomes in favour of event E is:

OPTIONS: (A) 4 (B) 6 (C) 10 (D) 26

ANSWER: (C) 10

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Probability of getting an irrational number at random from the numbers $\sqrt{3}$, $\sqrt{4}$, $3\sqrt{9}$, $3\sqrt{8}$, $\sqrt{5}$, 0, $4^{(3/2)}$ is:

OPTIONS: (A) 0 (B) $4/7$ (C) $3/7$ (D) 1

ANSWER: (C) $3/7$

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): The probability of a certain event E is 1. Reason (R): The sum of probabilities of all elementary events of an experiment is 1.

OPTIONS: (A) Both A and R true, R explains A (B) Both A and R true, R does not explain A (C) A true, R false (D) A false, R true

ANSWER: (A) Both A and R true, R explains A

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): The events "getting 2" and "not getting 2" in a single throw of an unbiased die are not equally likely events. Reason (R): The probability of getting 2 in a single throw of an unbiased die is $1/6$.

OPTIONS: (A) Both A and R true, R explains A (B) Both A and R true, R does not explain A (C) A true, R false (D) A false, R true

ANSWER: (A) Both A and R true, R explains A

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): The probability of an event can not be $1/0.9$. Reason (R): $0 \leq P(E) \leq 1$ for an event E.

OPTIONS: (A) Both A and R true, R explains A (B) Both A and R true, R does not explain A (C) A true, R false (D) A false, R true

ANSWER: (A) Both A and R true, R explains A

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Assertion (A): From a bag containing 5 red balls, 2 white balls and 3 green balls, the probability of drawing a non-white ball is $4/5$. Reason (R): For any event E, $P(E) + P(\text{not } E) = 1$.

OPTIONS: (A) Both A and R true, R explains A (B) Both A and R true, R does not explain A (C) A true, R false (D) A false, R true

ANSWER: (A) Both A and R true, R explains A

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: A bag contains 10 red pens, out of which 3 are defective and 8 blue pens, out of which 4 are defective. A pen is taken out at random. Find the probability that (i) it is a defective pen (ii) it is a non-defective red pen.

OPTIONS: (A) (i) $7/18$, (ii) $7/18$ (B) (i) $7/18$, (ii) $7/18$ (C) (i) $7/18$, (ii) $7/18$ (D) (i) $7/18$, (ii) $7/18$

ANSWER: (A) (i) $7/18$, (ii) $7/18$

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: A box consists of 60 wall clocks, out of which 40 are good, 15 have minor defects and the remaining are broken. Find probability that (i) the box will be rejected (ii) the clock has minor defect.

OPTIONS: (A) (i) $1/12$, (ii) $1/4$ (B) (i) $1/12$, (ii) $1/4$ (C) (i) $1/12$, (ii) $1/4$ (D) (i) $1/12$, (ii) $1/4$

ANSWER: (A) (i) $1/12$, (ii) $1/4$

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: A square dart board is divided into four equal squares. If a dart hits APOS, player wins ₹100. If it hits BPOQ, player loses ₹50. If it hits OQCR, player loses ₹100. If it hits ORDS, player wins ₹25. Find probability that (i) the player loses money (ii) the player wins ₹100.

OPTIONS: (A) (i) $1/2$, (ii) $1/4$ (B) (i) $1/2$, (ii) $1/4$ (C) (i) $1/2$, (ii) $1/4$ (D) (i) $1/2$, (ii) $1/4$

ANSWER: (A) (i) $1/2$, (ii) $1/4$

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Find the probability that a number selected at random from the numbers 30, 31, 32, ..., 60 is (i) a prime number (ii) a multiple of 6.

OPTIONS: (A) (i) $7/31$, (ii) $6/31$ (B) (i) $7/31$, (ii) $6/31$ (C) (i) $7/31$, (ii) $6/31$ (D) (i) $7/31$, (ii) $6/31$

ANSWER: (A) (i) $7/31$, (ii) $6/31$

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: Slips of letters of the word 'BACKGROUND' are put in a bowl. One slip is picked up at random. Find probability that it is (i) a vowel (ii) present in the word 'BALL'.

OPTIONS: (A) (i) $\frac{3}{10}$, (ii) $\frac{1}{5}$ (B) (i) $\frac{3}{10}$, (ii) $\frac{1}{5}$ (C) (i) $\frac{3}{10}$, (ii) $\frac{1}{5}$ (D) (i) $\frac{3}{10}$, (ii) $\frac{1}{5}$

ANSWER: (A) (i) $\frac{3}{10}$, (ii) $\frac{1}{5}$

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A box contains 6 blue, 4 white and 8 red marbles. A marble is drawn at random. Find probability that it is (i) white (ii) white or red (iii) not red.

OPTIONS: (A) (i) $\frac{2}{9}$, (ii) $\frac{2}{3}$, (iii) $\frac{5}{9}$ (B) (i) $\frac{2}{9}$, (ii) $\frac{2}{3}$, (iii) $\frac{5}{9}$ (C) (i) $\frac{2}{9}$, (ii) $\frac{2}{3}$, (iii) $\frac{5}{9}$ (D) (i) $\frac{2}{9}$, (ii) $\frac{2}{3}$, (iii) $\frac{5}{9}$

ANSWER: (A) (i) $\frac{2}{9}$, (ii) $\frac{2}{3}$, (iii) $\frac{5}{9}$

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A game of chance consists of spinning a wheel which comes to rest at one of the numbers from 1 to 10 with equal probabilities. What is the probability that the wheel stops at (i) a prime number greater than 2 (ii) an odd number less than 9 (iii) a multiple of 4?

OPTIONS: (A) (i) $\frac{3}{10}$, (ii) $\frac{2}{5}$, (iii) $\frac{1}{5}$ (B) (i) $\frac{3}{10}$, (ii) $\frac{2}{5}$, (iii) $\frac{1}{5}$ (C) (i) $\frac{3}{10}$, (ii) $\frac{2}{5}$, (iii) $\frac{1}{5}$ (D) (i) $\frac{3}{10}$, (ii) $\frac{2}{5}$, (iii) $\frac{1}{5}$

ANSWER: (A) (i) $\frac{3}{10}$, (ii) $\frac{2}{5}$, (iii) $\frac{1}{5}$

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Two dice are rolled together. Find the probability that (i) the sum of the numbers obtained is 10 (ii) the product of the numbers obtained is 6.

OPTIONS: (A) (i) $\frac{1}{12}$, (ii) $\frac{1}{9}$ (B) (i) $\frac{1}{12}$, (ii) $\frac{1}{9}$ (C) (i) $\frac{1}{12}$, (ii) $\frac{1}{9}$ (D) (i) $\frac{1}{12}$, (ii) $\frac{1}{9}$

ANSWER: (A) (i) $\frac{1}{12}$, (ii) $\frac{1}{9}$

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Two dice are rolled together. Find the probability that at least one of the numbers obtained is a multiple of 3.

OPTIONS: (A) $\frac{5}{9}$ (B) $\frac{5}{9}$ (C) $\frac{5}{9}$ (D) $\frac{5}{9}$

ANSWER: (A) $\frac{5}{9}$

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A lot consists of 200 pens of which 180 are good and the rest are defective. A customer will buy a pen if it is not defective. What is the probability that the customer will not buy it? Another lot of 100 pens containing 80 good pens is mixed. Now what is the probability that the customer will buy the pen?

OPTIONS: (A) (i) $\frac{1}{10}$, (ii) $\frac{13}{15}$ (B) (i) $\frac{1}{10}$, (ii) $\frac{13}{15}$ (C) (i) $\frac{1}{10}$, (ii) $\frac{13}{15}$ (D) (i) $\frac{1}{10}$, (ii) $\frac{13}{15}$

ANSWER: (A) (i) $\frac{1}{10}$, (ii) $\frac{13}{15}$

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A game of chance consists of spinning an arrow which comes to rest in one of the six equal sectors. A person spins an arrow once. What is the probability that the person (i) wins a laptop (ii) does not get a chance to spin again?

OPTIONS: (A) (i) $\frac{1}{6}$, (ii) $\frac{5}{6}$ (B) (i) $\frac{1}{6}$, (ii) $\frac{5}{6}$ (C) (i) $\frac{1}{6}$, (ii) $\frac{5}{6}$ (D) (i) $\frac{1}{6}$, (ii) $\frac{5}{6}$

ANSWER: (A) (i) $\frac{1}{6}$, (ii) $\frac{5}{6}$

YEAR: 2026

CHAPTER: 14 - PROBABILITY

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A bag contains 10 red pens, out of which 3 are defective and 8 blue pens, out of which 4 are defective. A pen is taken out at random. Find the probability that (i) it is a defective pen (ii) it is a non-defective red pen.

OPTIONS: (A) (i) $\frac{7}{18}$, (ii) $\frac{7}{18}$ (B) (i) $\frac{7}{18}$, (ii) $\frac{7}{18}$ (C) (i) $\frac{7}{18}$, (ii) $\frac{7}{18}$ (D) (i) $\frac{7}{18}$, (ii) $\frac{7}{18}$

ANSWER: (A) (i) $\frac{7}{18}$, (ii) $\frac{7}{18}$
